

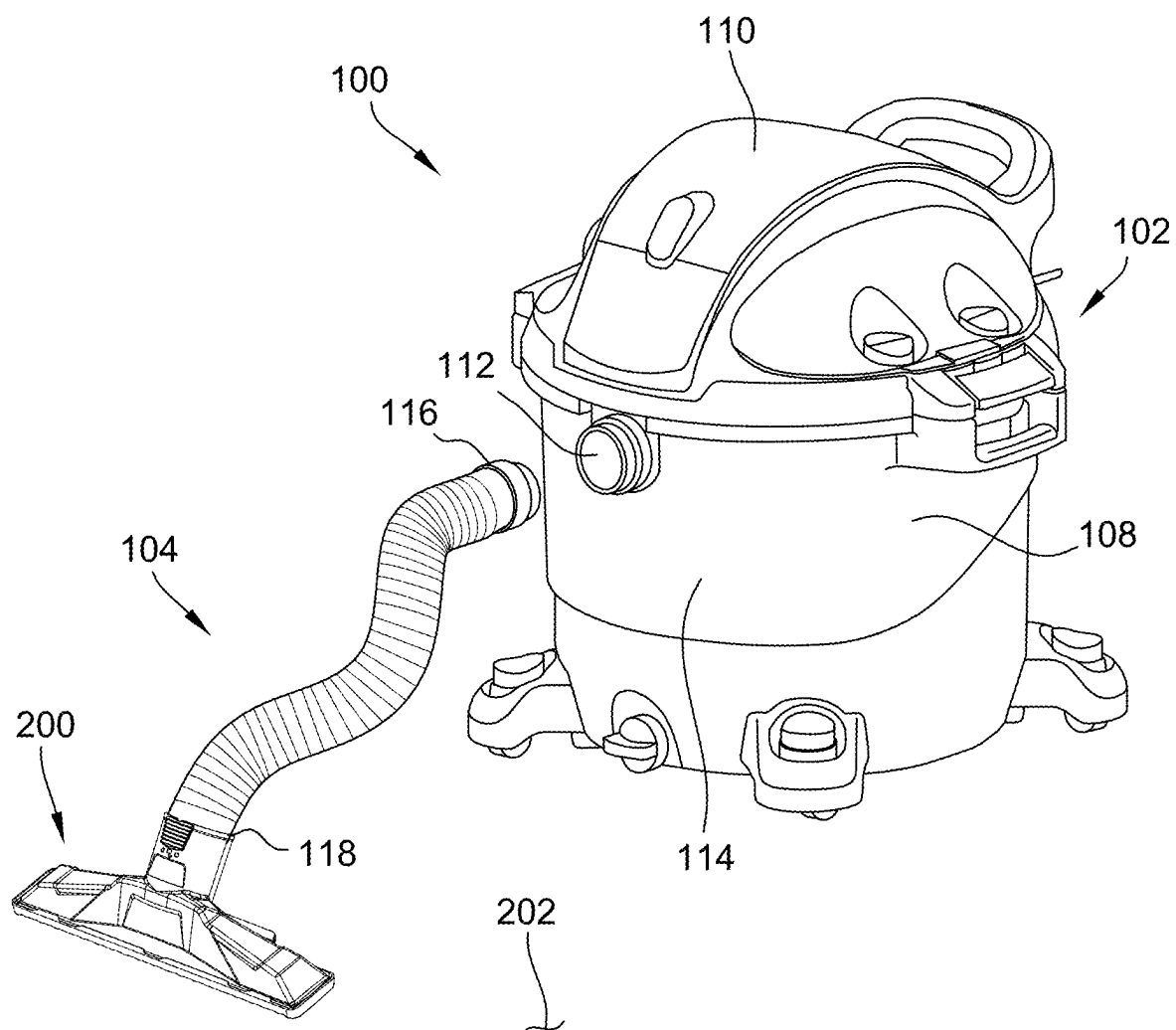


FIG. 1

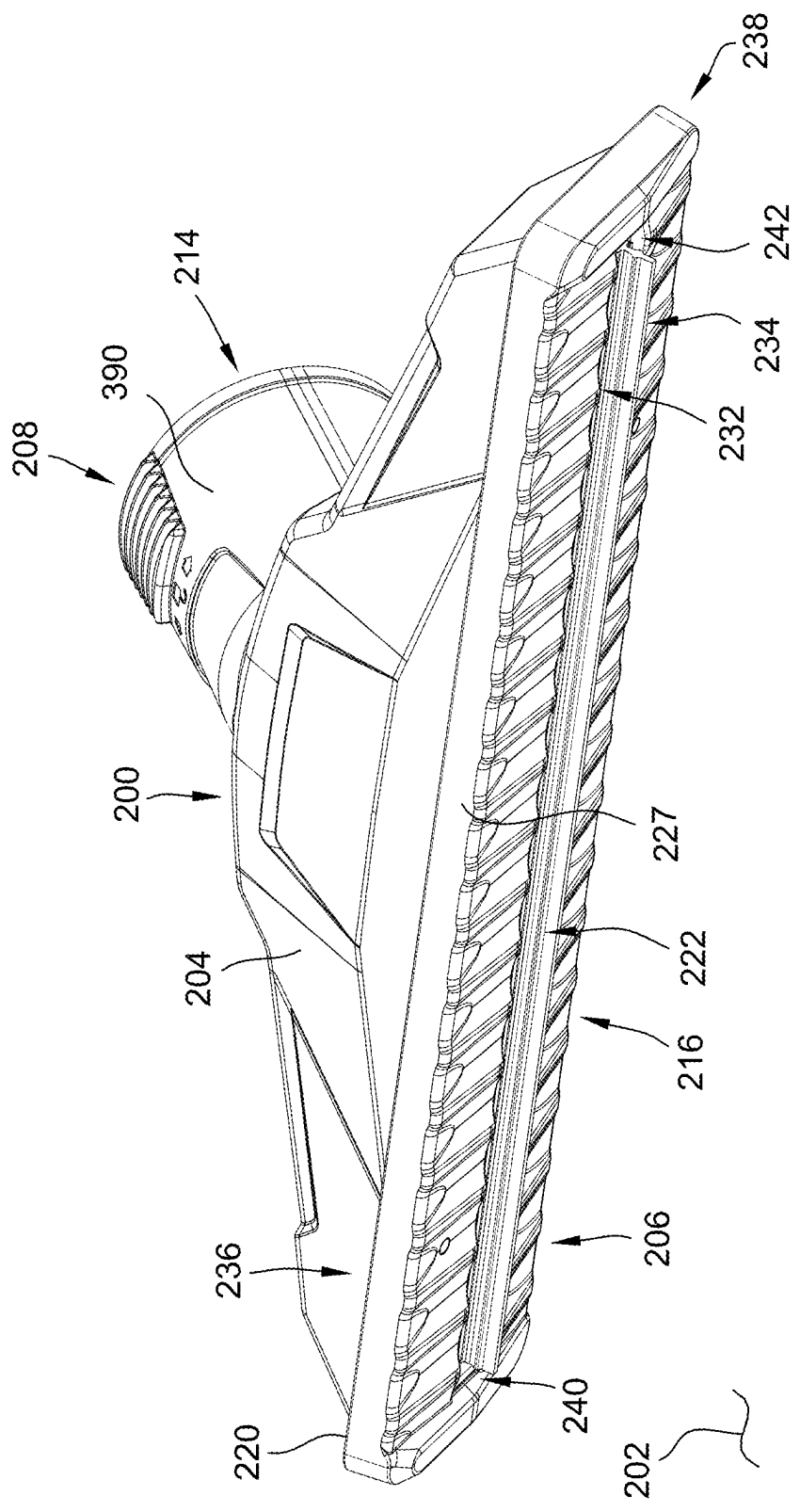
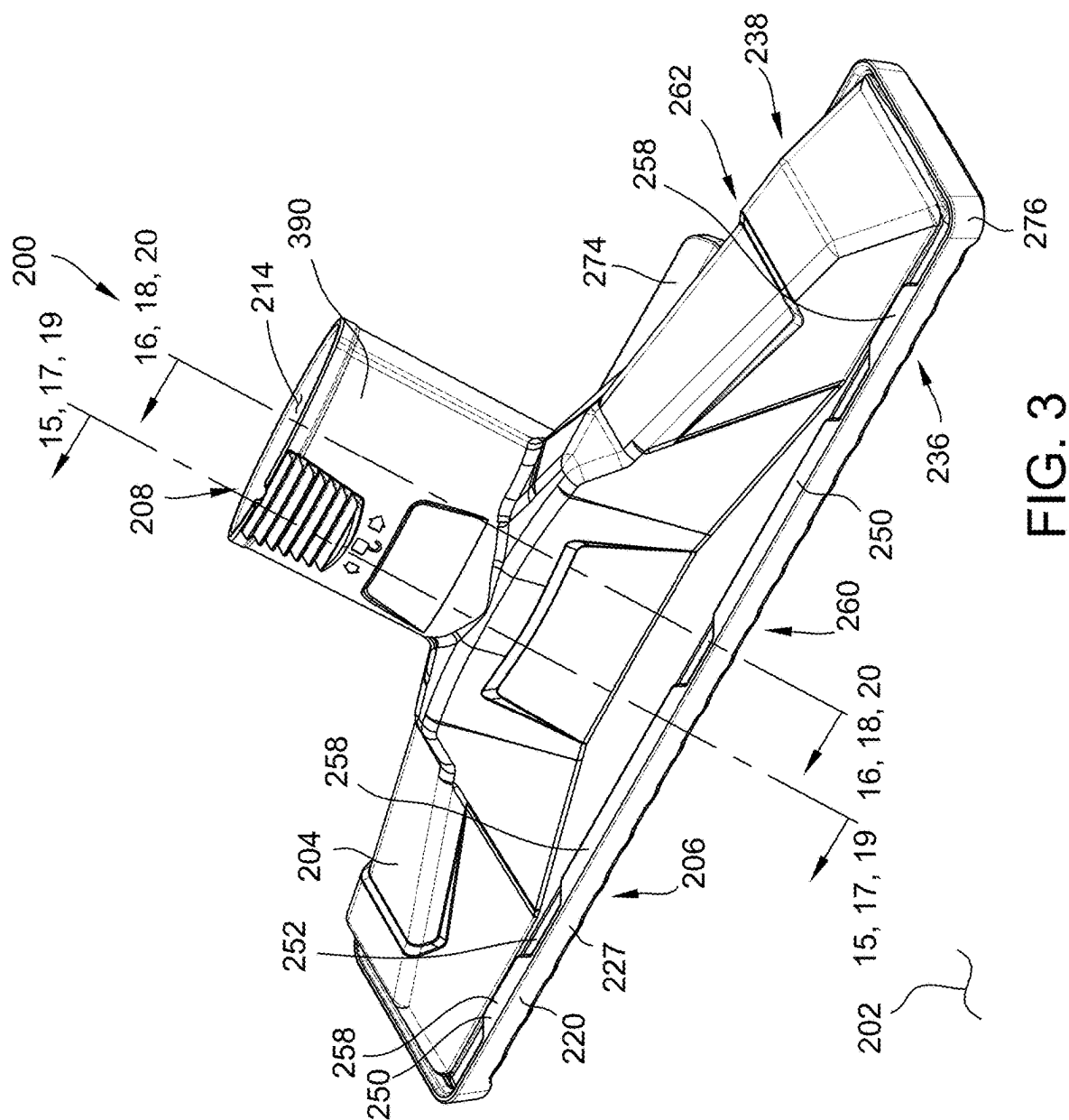


FIG. 2



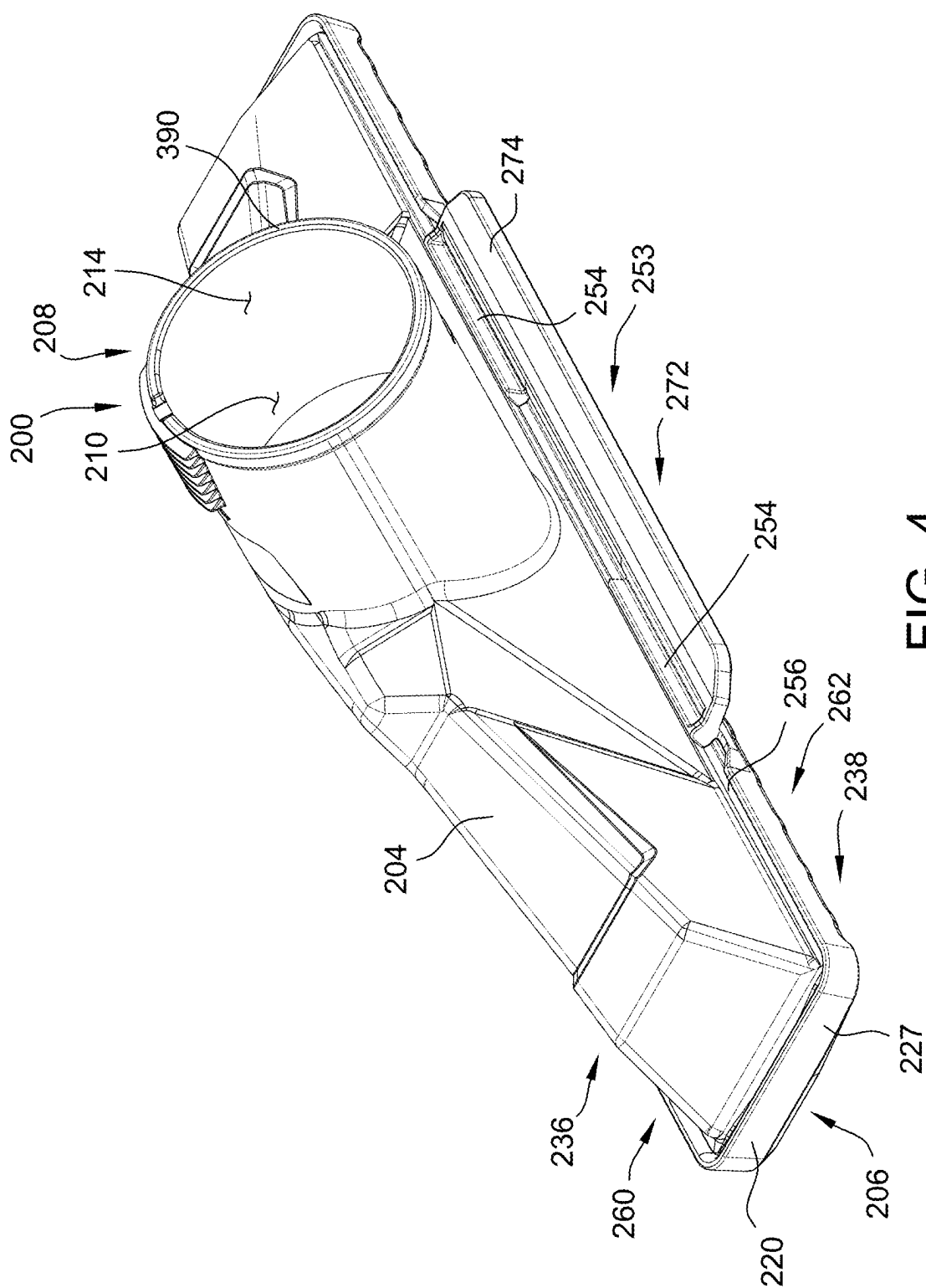


FIG. 4

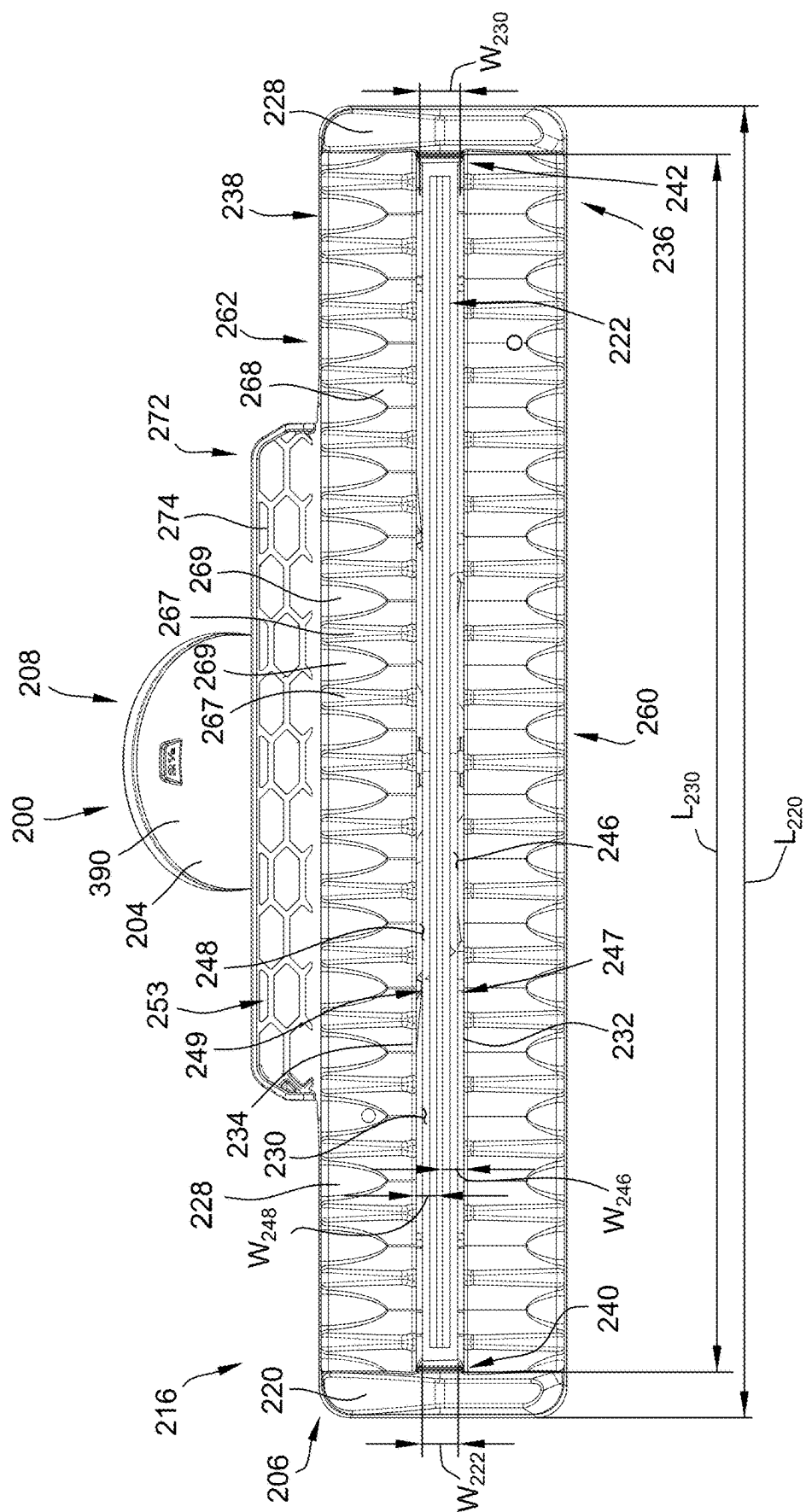


Fig. 5

6
G.
F

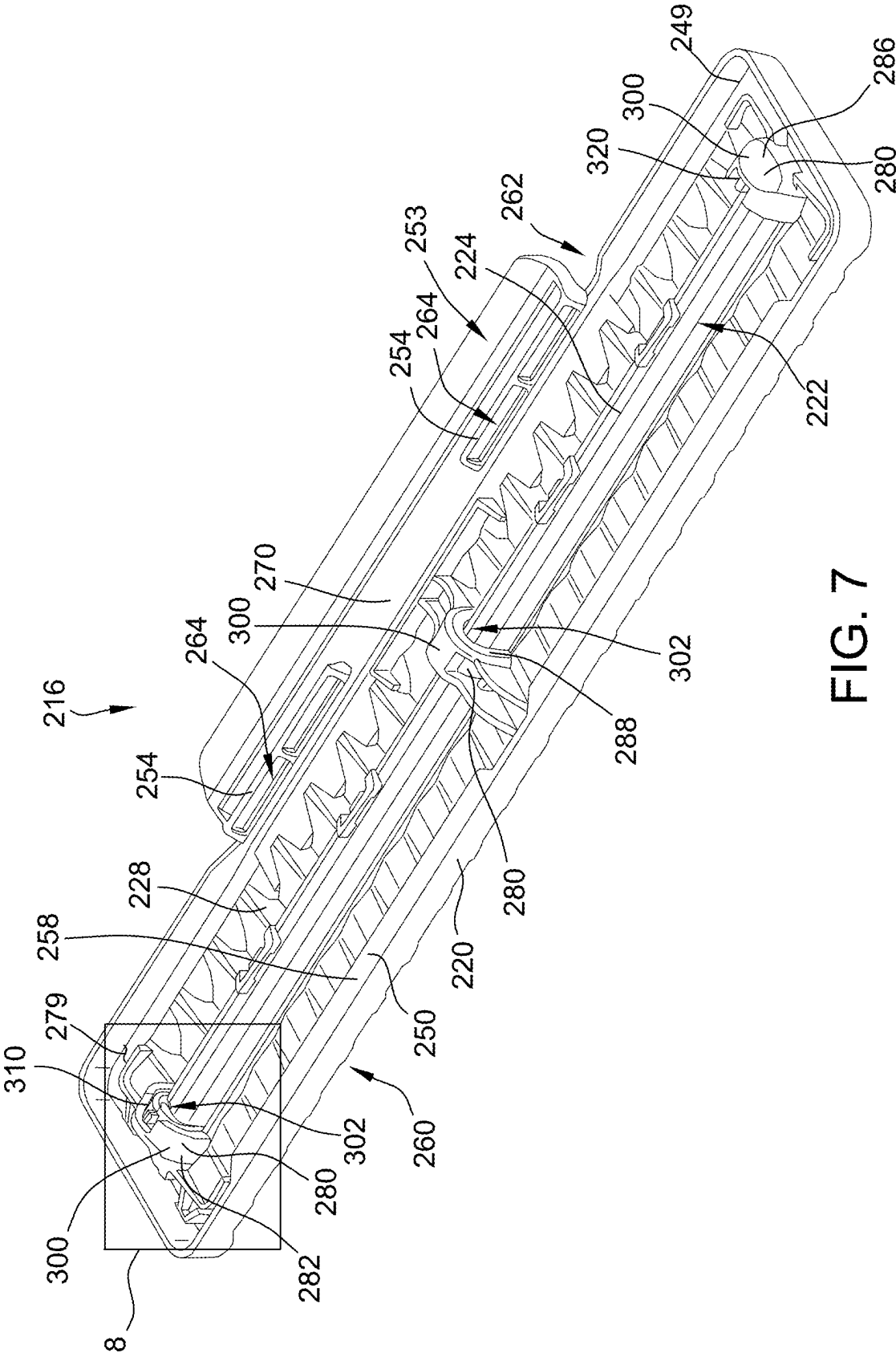


FIG. 7

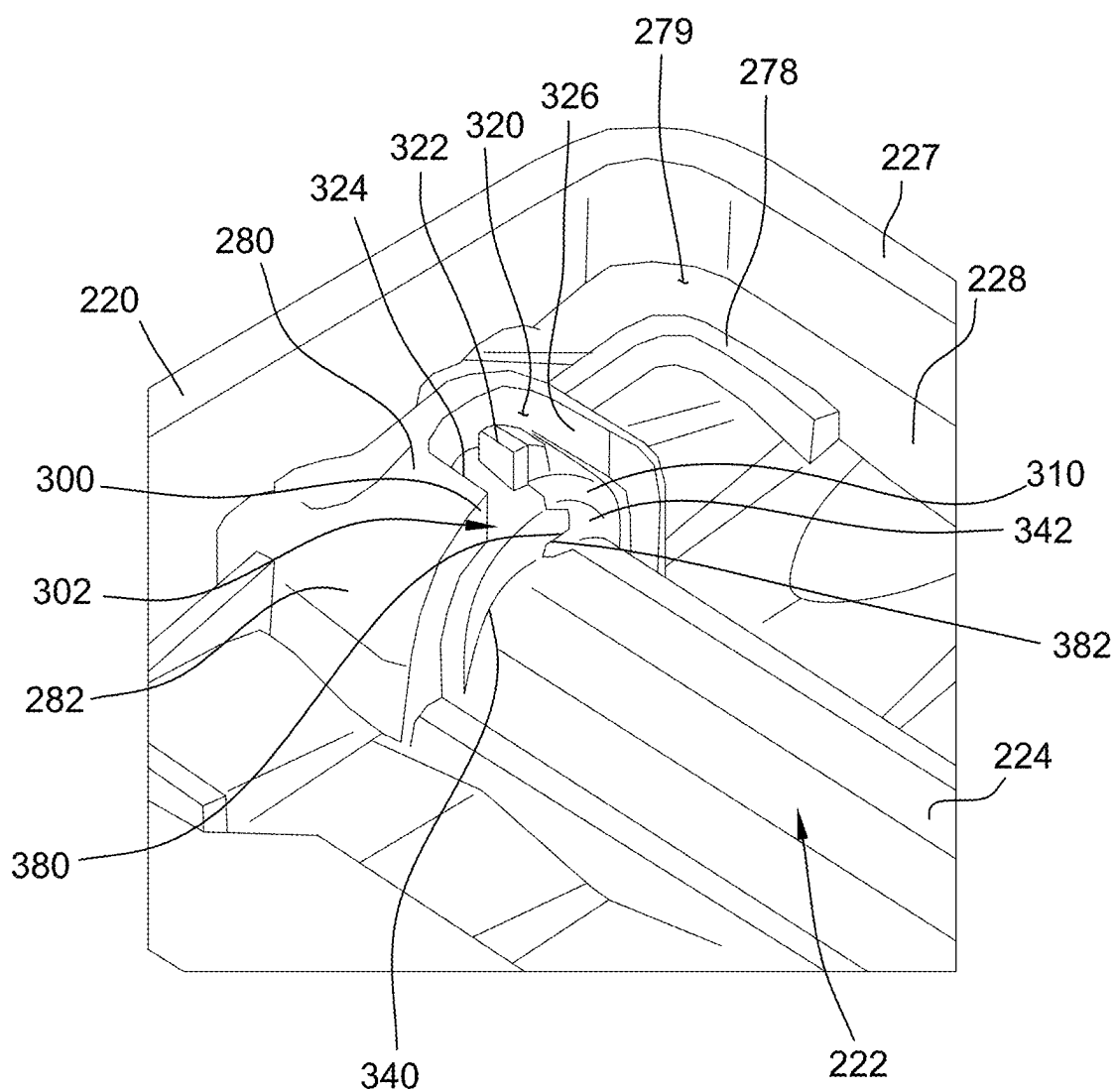


FIG. 8

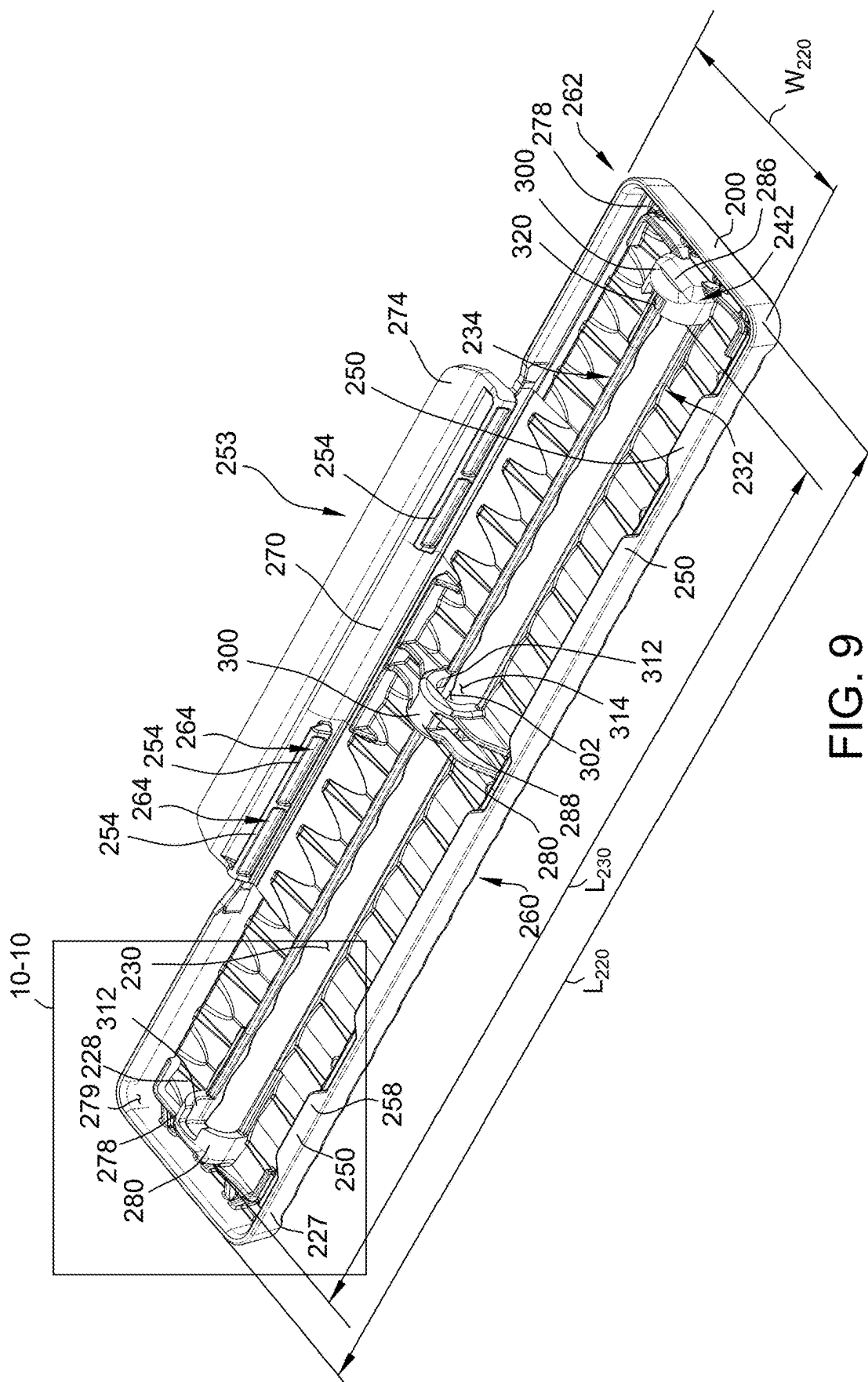


FIG. 9

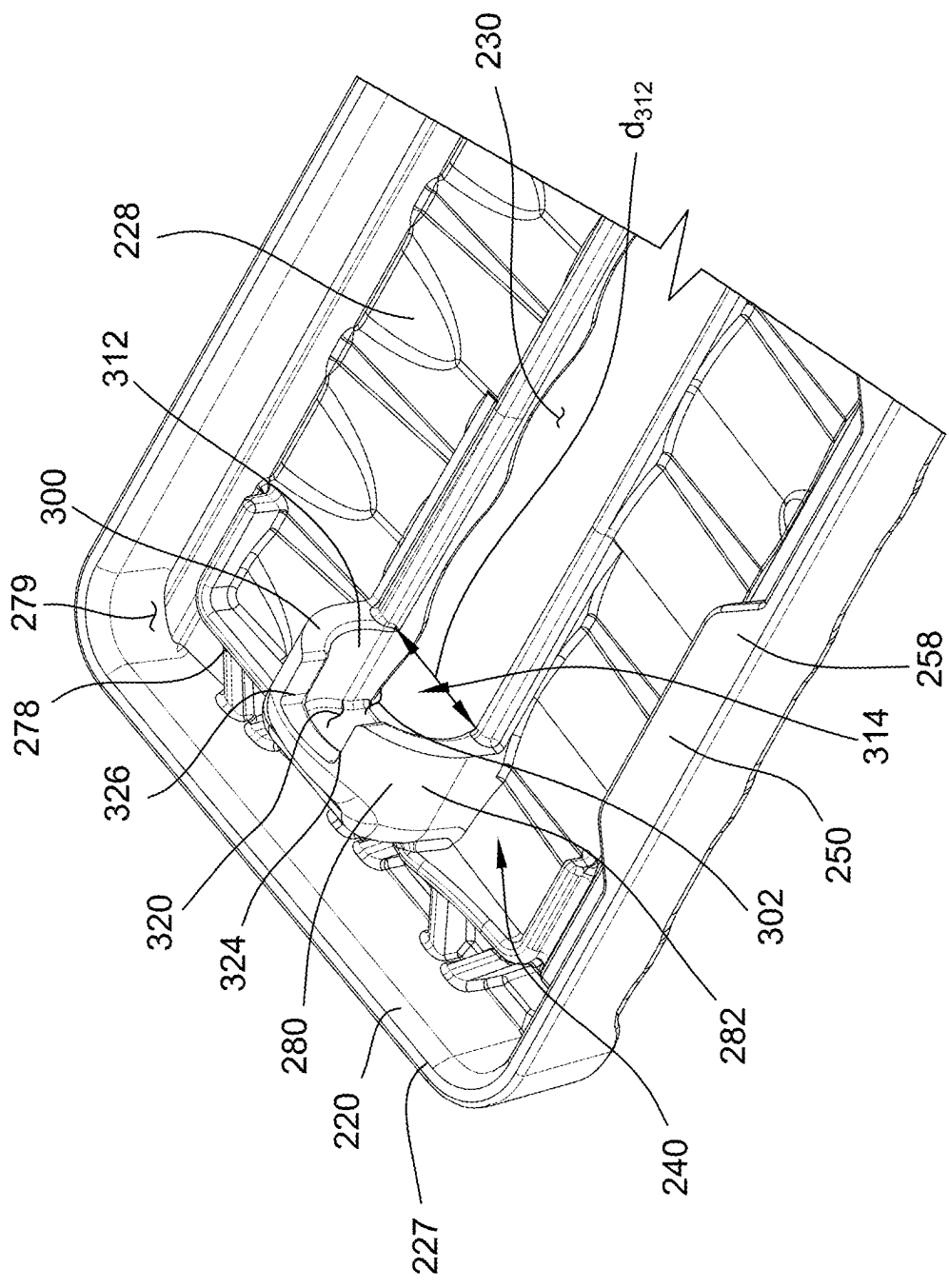


FIG. 10

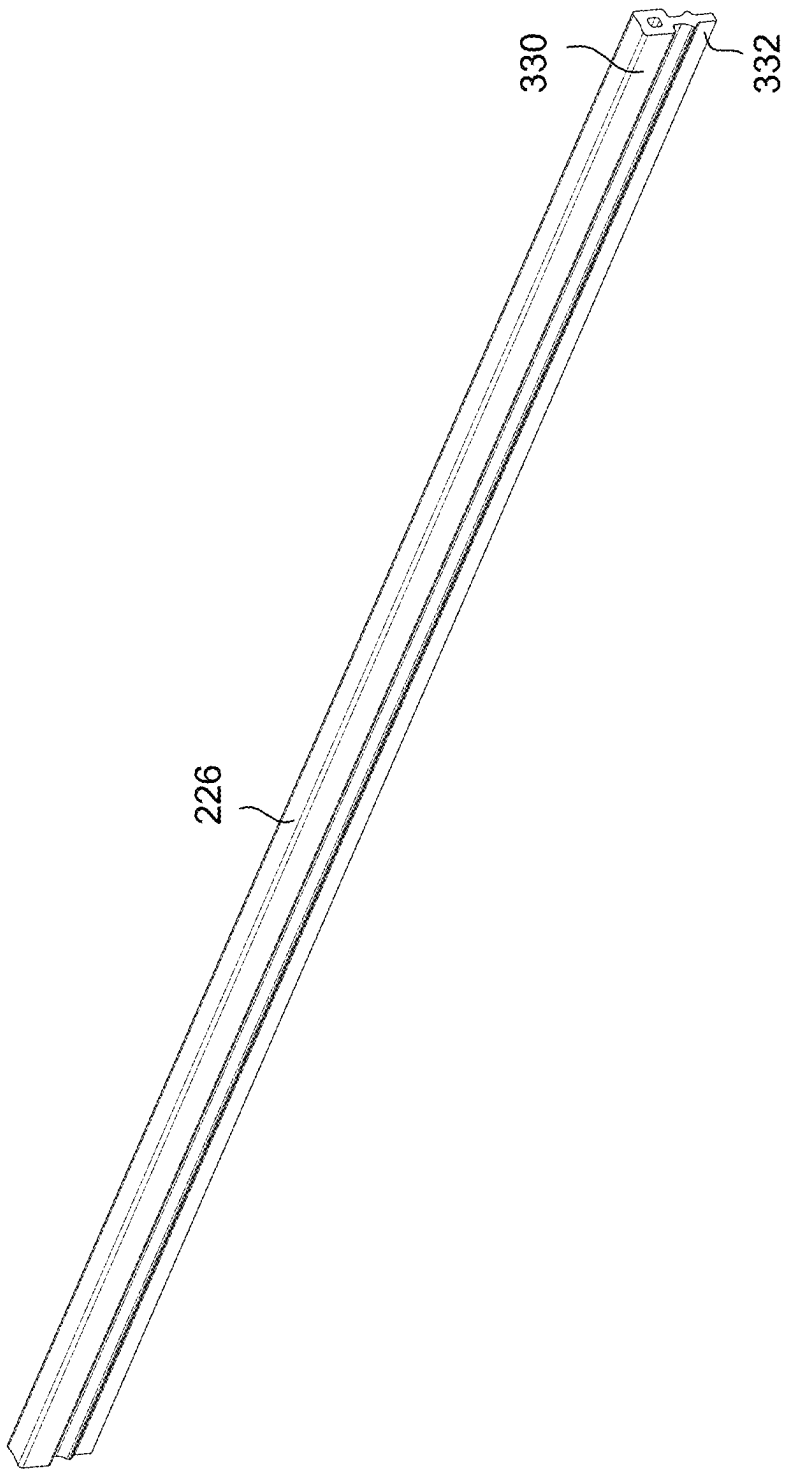


FIG. 12

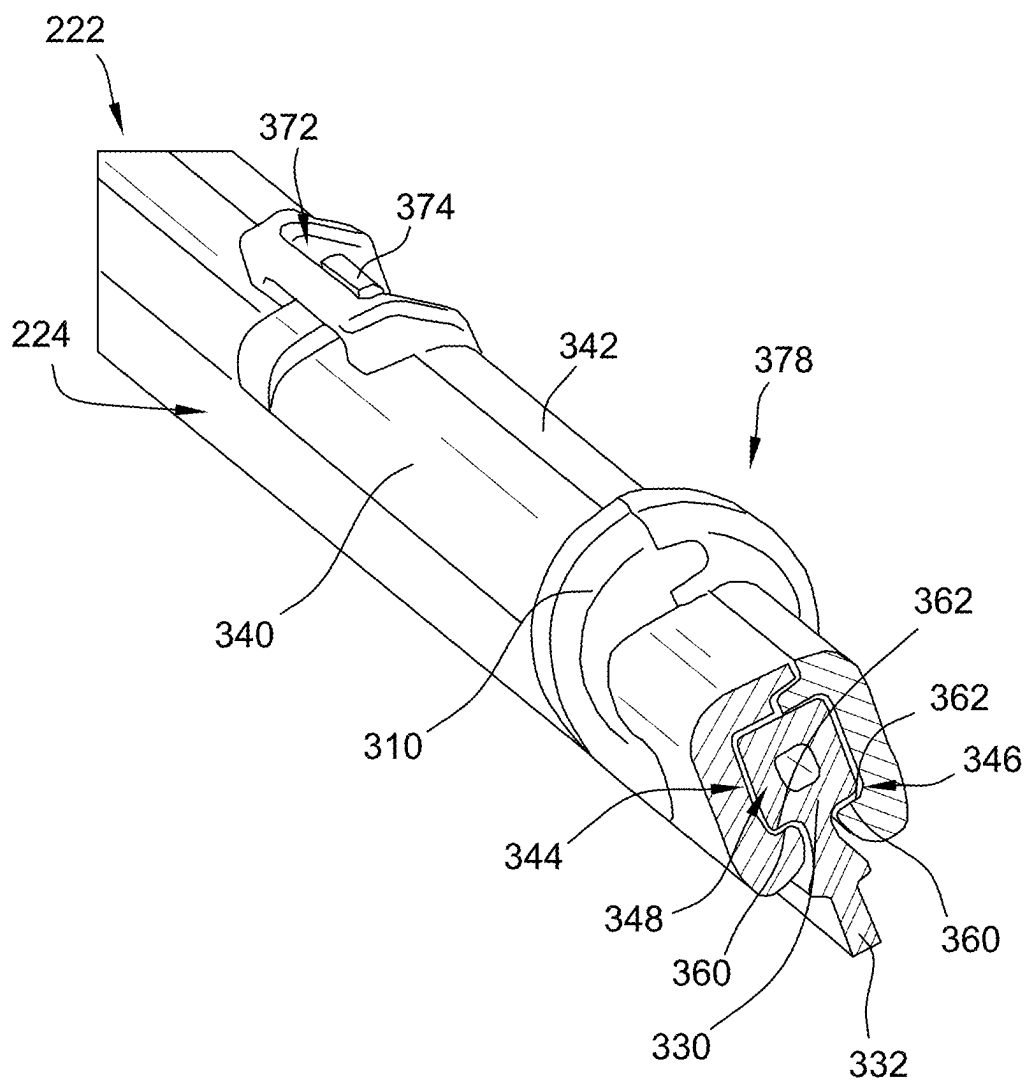


FIG. 13

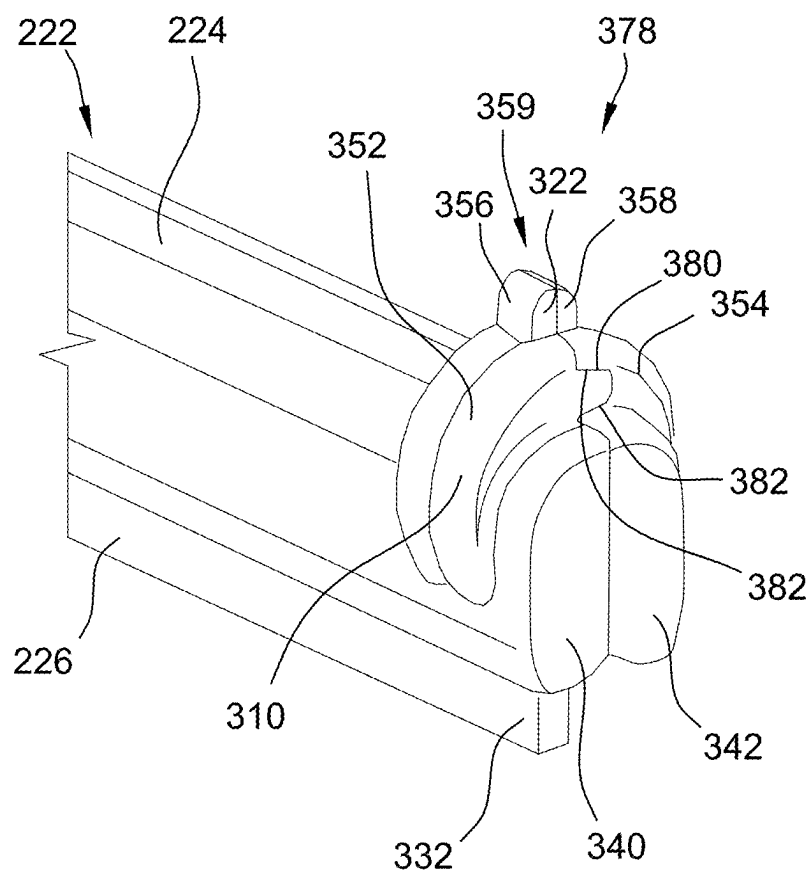


FIG. 14

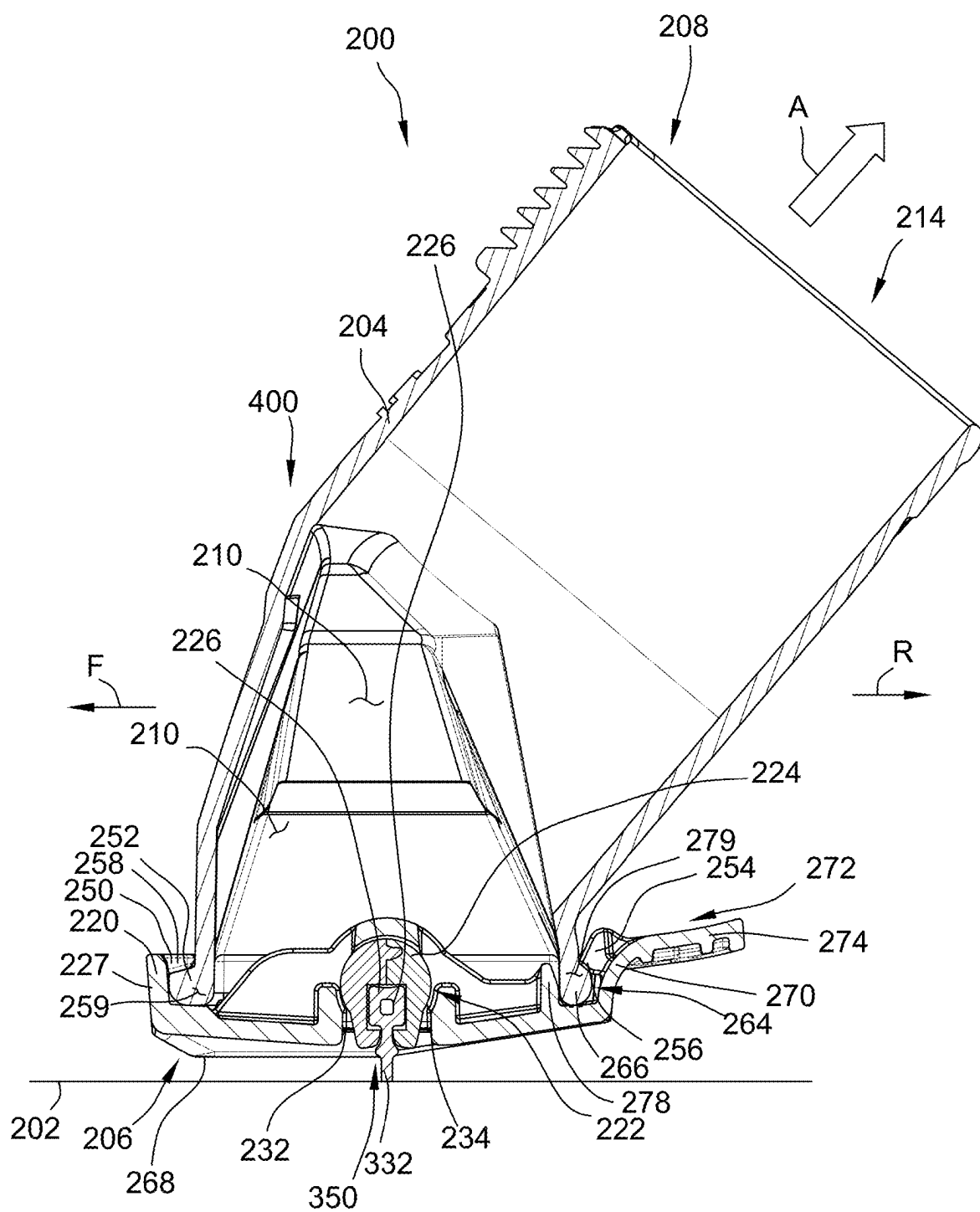


FIG. 15

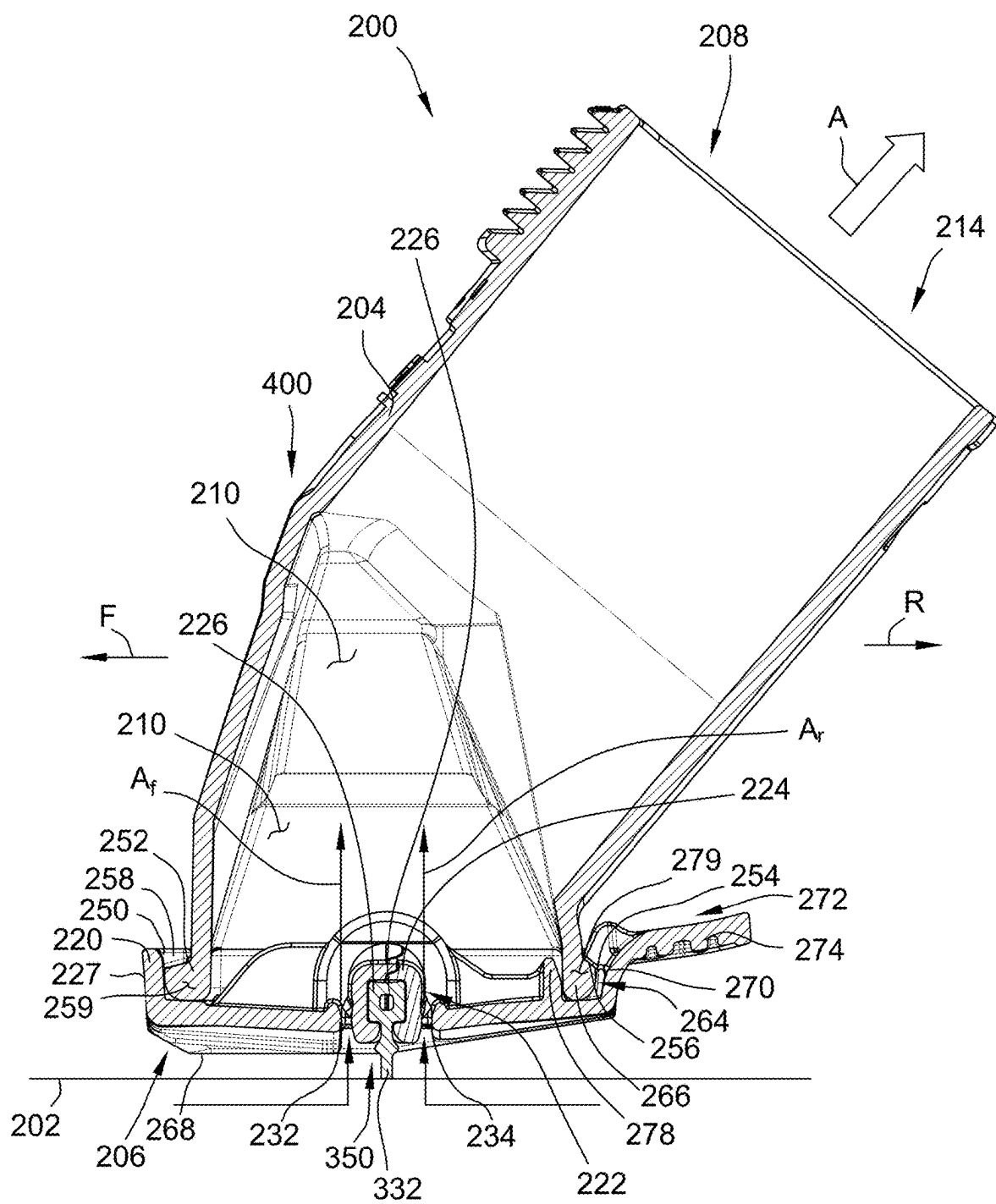


FIG. 16

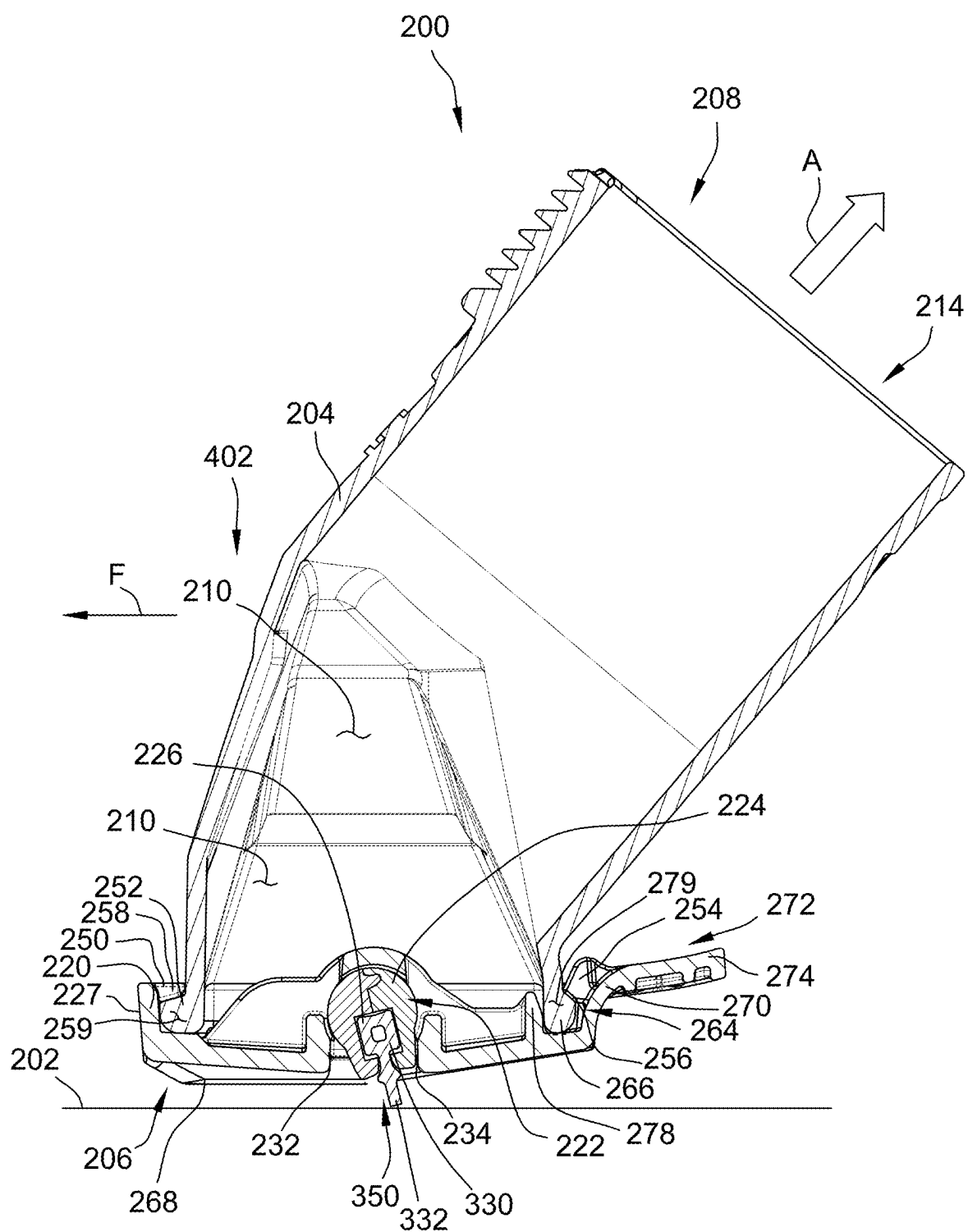


FIG. 17

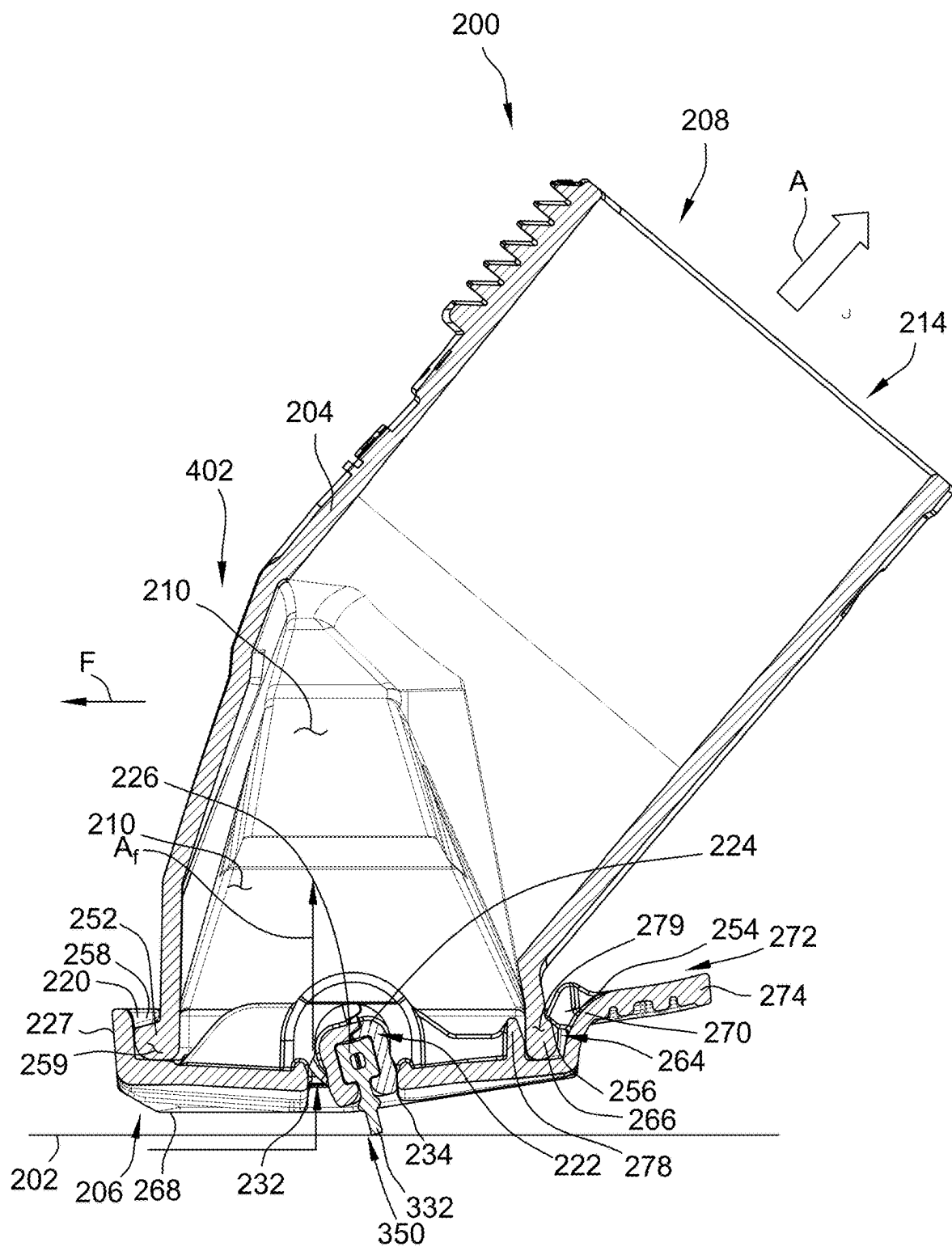


FIG. 18

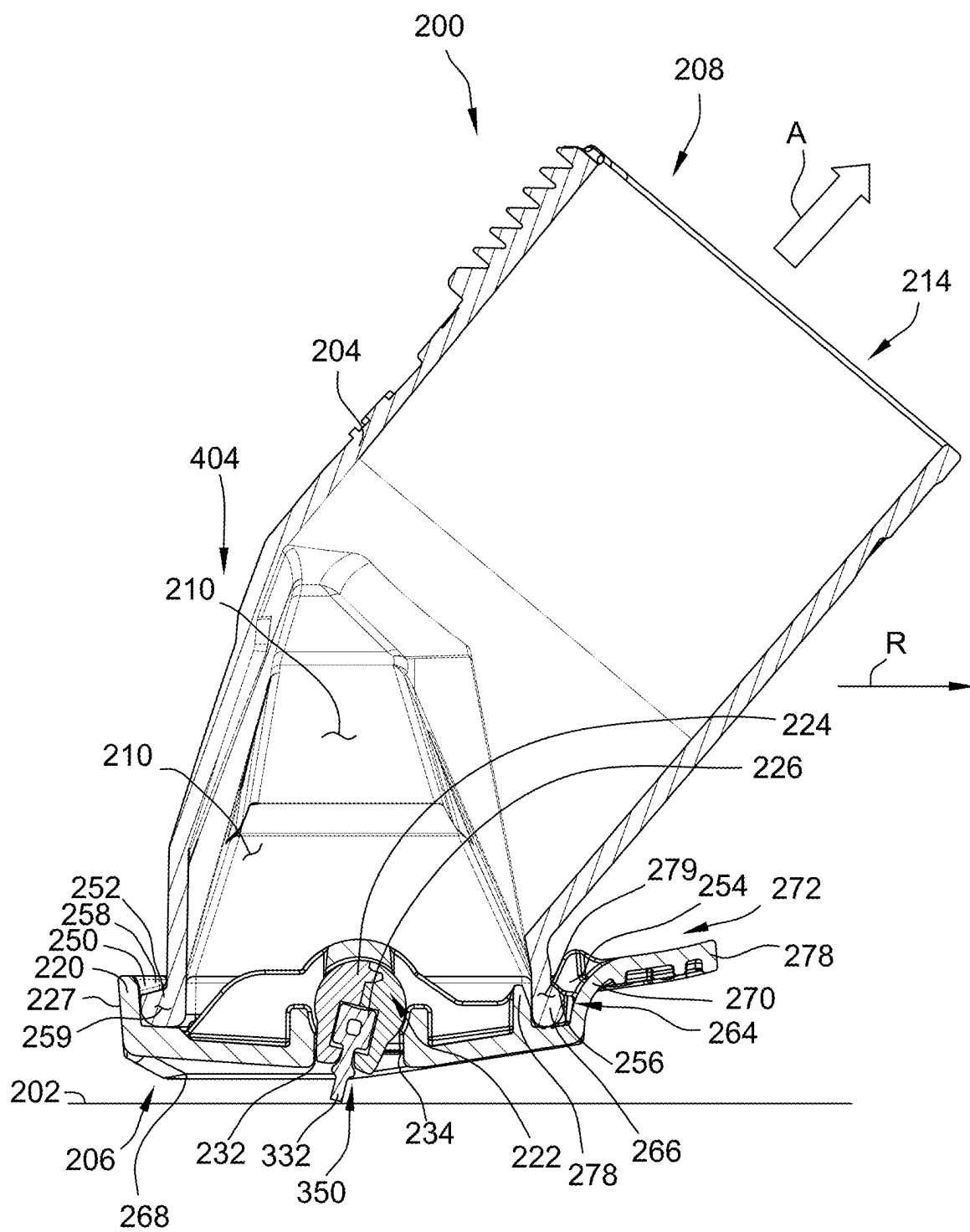


FIG. 19

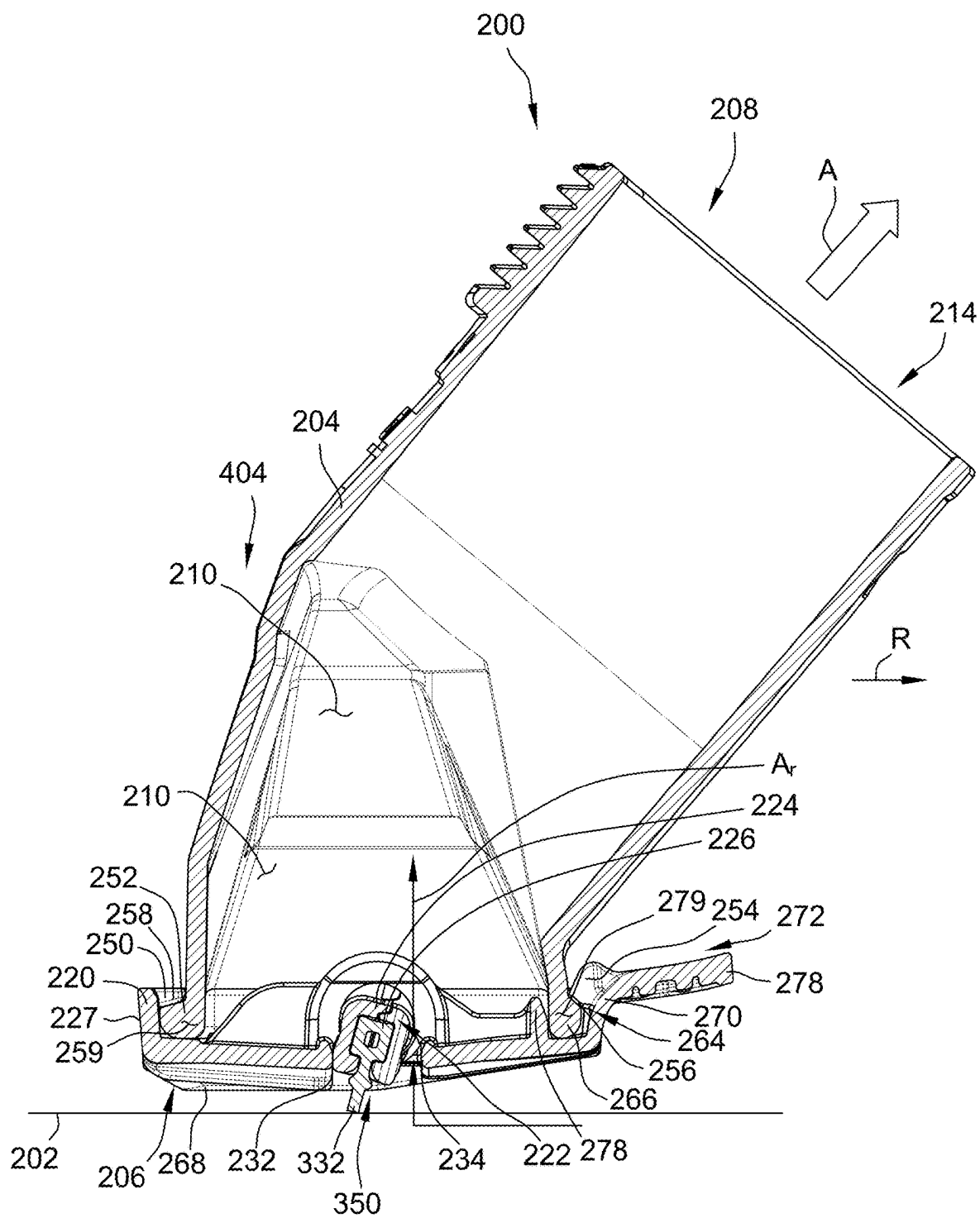


FIG. 20

SQUEEGEE ATTACHMENT ASSEMBLY AND VACUUM ACCESSORY INCLUDING SAME

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to India patent application Ser. No. 202411015366, filed Mar. 1, 2024, the disclosure of which is hereby incorporated by reference in its entirety.

FIELD

[0002] The field of the disclosure relates generally to vacuum accessories and attachments and, more particularly, to squeegee attachments for use with a wet/dry vacuum cleaner.

BACKGROUND

[0003] Vacuum cleaners, such as wet/dry vacuums, typically include a suction unit connected to a conduit (e.g., a flexible hose or tube). Various attachment tools or accessories are available that may be connected to the conduit, such as brushes, squeegees, and/or crevice tools.

[0004] Conventional squeegee tools, for use with a vacuum, include a squeegee blade fixed relative to the suction inlet such that the squeegee blade may be used to squeegee, e.g., collect material such as liquid or debris from a surface, while the suction inlet simultaneously aspirates the material. When the squeegee is fixed to one side of the suction inlet, the squeegee tool may only be used in one direction. For example, when the squeegee is positioned in front of the suction inlet the user can only draw the squeegee tool in a backwards motion to simultaneously aspirate the material collected by the squeegee. Moreover, when the squeegee is fixed relative to the suction inlet, forces exerted on the squeegee during operation (e.g., to flex the squeegee) may cause excessive wear on the squeegee and/or skipping or “chattering” of the squeegee across the surface being cleaned.

[0005] This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

SUMMARY

[0006] In one aspect, a vacuum accessory includes a nozzle body defining a cavity and a vacuum inlet port selectively connectable to a vacuum conduit. The vacuum accessory also includes a squeegee attachment assembly removably connected to the nozzle body. The squeegee attachment assembly includes a housing that is selectively connectable to the nozzle body and defines a suction inlet, and a squeegee assembly rotatably connected to the housing such that the squeegee assembly is rotatable relative to the suction inlet. The squeegee assembly is at least partially disposed within the suction inlet. The squeegee assembly is rotatable between a forward position, in which the squeegee assembly occludes a forward side of the suction inlet, and a

rearward position, in which the squeegee assembly occludes a rearward side of the suction inlet.

[0007] In another aspect, a squeegee attachment assembly for use with a vacuum accessory includes a housing that defines a suction inlet and includes at least one connector to selectively connect the squeegee attachment assembly to a nozzle body. The squeegee attachment assembly further includes a squeegee assembly rotatably connected to the housing such that the squeegee assembly is rotatable relative to the suction inlet. The squeegee assembly is at least partially disposed within the suction inlet, and is rotatable between a forward position, in which the squeegee assembly occludes a forward side of the suction inlet, and a rearward position, in which the squeegee assembly occludes a rearward side of the suction inlet.

[0008] In another aspect, a method of assembling a vacuum accessory including a nozzle body and a squeegee attachment assembly is provided. The squeegee attachment assembly includes a squeegee assembly and a housing defining a suction inlet. The method includes positioning a base portion of a squeegee blade in a holder pocket defined by one of a first holder bracket and a second holder bracket of a holder assembly and connecting the first holder bracket to the second holder bracket to retain the base portion within the holder assembly. The method includes rotatably connecting the holder assembly to a clip of the housing such that the squeegee assembly is at least partially disposed within the suction inlet and rotatable relative to the suction inlet between a forward position, in which the squeegee assembly occludes a front side of the suction inlet, and a rearward position, in which the squeegee assembly occludes a rear side of the suction inlet and connecting the squeegee assembly to the nozzle body.

[0009] Various refinements exist of the features noted in relation to the above-mentioned aspects of the present disclosure. Further features may also be incorporated in the above-mentioned aspects of the present disclosure as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to any of the illustrated embodiments of the present disclosure may be incorporated into any of the above-described aspects of the present disclosure, alone or in any combination.

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a perspective view of an example vacuum cleaning system including an example vacuum accessory and vacuum, illustrated as a wet/dry vacuum cleaner.

[0011] FIG. 2 is a lower front perspective view of the example vacuum accessory shown in FIG. 1, including a nozzle body and a squeegee attachment assembly.

[0012] FIG. 3 is an upper front perspective view of the vacuum accessory shown in FIG. 2.

[0013] FIG. 4 is a rear perspective view of the vacuum accessory shown in FIG. 2.

[0014] FIG. 5 is a bottom view of the vacuum accessory shown in FIG. 2.

[0015] FIG. 6 is a lower perspective view of the nozzle body of the vacuum accessory shown in FIG. 2.

[0016] FIG. 7 is an upper perspective view of the squeegee attachment assembly of the vacuum accessory shown in FIG. 2.

[0017] FIG. 8 is an enlarged view of a portion of the squeegee attachment assembly shown in FIG. 7.

[0018] FIG. 9 is another upper perspective view of the squeegee attachment assembly with a squeegee assembly thereof omitted.

[0019] FIG. 10 is an enlarged view of a portion of the squeegee attachment assembly shown in FIG. 9.

[0020] FIG. 11 is a perspective view of a squeegee assembly of the squeegee attachment assembly, including a holder and a squeegee blade.

[0021] FIG. 12 is a perspective view of the squeegee blade of the squeegee assembly of FIG. 11.

[0022] FIG. 13 is an enlarged cut-away view of a portion of the squeegee assembly of FIG. 11, illustrating attachment features and a rotational bearing element of the squeegee assembly.

[0023] FIG. 14 is another enlarged view of a portion of the squeegee assembly.

[0024] FIG. 15 is a cross-sectional view of the vacuum accessory, taken along line 15-15 in FIG. 3, illustrating the squeegee assembly in a neutral position.

[0025] FIG. 16 is another cross-sectional view of the vacuum accessory taken along line 16-16 in FIG. 3, illustrating the squeegee assembly in the neutral position.

[0026] FIG. 17 is a cross-sectional view of the vacuum accessory, taken along line 17-17 in FIG. 3, illustrating the squeegee assembly in a rearward position.

[0027] FIG. 18 is another cross-sectional view of the vacuum accessory, taken along line 18-18 in FIG. 3, with the squeegee assembly in a rearward position.

[0028] FIG. 19 is a cross-sectional view of the vacuum accessory, taken along line 19-19 in FIG. 3, with the squeegee assembly in a forward position.

[0029] FIG. 20 is another cross-sectional view of the vacuum accessory, taken along line 20-20 in FIG. 3, with the squeegee assembly in the forward position.

[0030] Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

[0031] Embodiments of the vacuum accessories and squeegee attachment assemblies described herein facilitate improved vacuuming and squeegeeing of a surface, such as a floor or a window. Embodiments of the disclosed vacuum accessories include a hollow nozzle body and a squeegee attachment assembly that is removably connectable to the nozzle body. The squeegee attachment assembly includes a housing defining a suction inlet for suctioning or aspirating material, e.g., liquids and/or debris, and a squeegee assembly that includes a squeegee blade for squeegeeing the surface to gather material to be drawn into the suction inlet.

[0032] The squeegee assembly is rotatably connected to a housing and is disposed generally in the middle of the suction inlet in some embodiments, dividing the suction inlet into a forward side and an opposing rear side. The squeegee assembly may be rotated into a forward position, where at least a forward side of the suction inlet is occluded by the squeegee assembly while a rear side of the suction inlet is unblocked. Similarly, the squeegee assembly may be rotated into a rearward position, where the rear side of the suction inlet is occluded by the squeegee assembly while the front side of the suction inlet is unblocked. As such, in the forward position or rearward position, the squeegee attachment blocks a portion of the suction inlet, narrowing the area of the suction inlet thereby increasing the suction force. In embodiments described herein, the squeegee attachment

assembly may be operated in either a forward direction (e.g., where the squeegee assembly is pushed away from an operator), in which the squeegee assembly is rotated into the rearward position, or an opposing rearward direction (e.g., where the squeegee assembly is pulled towards the operator), in which the squeegee assembly is rotated into the forward position. The forward and rearward positions may be considered end positions such that the squeegee assembly is restricted from rotating beyond the forward and rearward positions.

[0033] The rotational connection of the squeegee assembly provides ease of use by reducing the amount of force a user must impart to maintain contact between the squeegee blade and a surface, e.g., a floor. The rotational connection of the squeegee assembly allows for greater material selection for the squeegee blade because the squeegee blade experiences lower frictional forces and does not need to be as flexible as compared to other squeegee blades, for example, that bend or flex to seal off portion of a suction inlet. Rather, the rotational connection is used to reposition the blade relative to a surface. As the blade does not need to be flexed or bent into position, and the amount of force a user imparts is reduced, embodiments described herein have reduction in wear fatigue of the squeegee blade, increasing the blades useable lifespan.

[0034] In embodiments described herein, contact between the squeegee blade and the surface automatically rotates the squeegee blade from the forward position to the rearward position when the squeegee attachment assembly is transitioned between a forward motion to a rearward motion relative to a surface. Similarly, contact between the squeegee blade and the surface automatically rotates the squeegee blade from the rearward position to the forward position when the squeegee attachment assembly is transitioned between a rearward motion to a forward motion relative to a surface.

[0035] In embodiments described herein, a bottom surface of the accessory includes serrations may induce turbulent flow leading towards the squeegee blade and/or the inlet, reducing whistling noise generated by the squeegee attachment assembly. Narrowing of the suction inlet, e.g., when the squeegee assembly is in the forward or rearward positions, increases inlet flow velocities which may generate harmonic whistling. The turbulent flow created by the serrations inhibits the formation of standing harmonics preventing or reducing this harmonic whistling. Furthermore, the serrations distribute inlet flow evenly along the entire length of the suction inlet preventing localized high velocity inlet flow, e.g., at the middle of the suction inlet.

[0036] Embodiments of the squeegee attachment assemblies described herein include one or more components that may be assembled by hand and without the use of additional fasteners, e.g., screws or bolts, and without the use of additional tools. For example, in some embodiments, a squeegee assembly includes a squeegee holder having attachment features for connecting the two pieces together around a portion of the base of the squeegee blade to retain the squeegee blade in the holder. In another example, the holder may be press fit into a pocket defined by a clip formed on a housing of the squeegee attachment assembly to rotationally connect the squeegee assembly to the housing of squeegee attachment assembly. In certain embodiments, the squeegee attachment assembly includes three components—

the squeegee blade, the squeegee holder, and the housing, which may be assembled together without the use of tools and/or additional fasteners.

[0037] FIG. 1 is a perspective view of an example vacuum cleaning system **100** that includes a vacuum accessory **200** suitable for use with a vacuum **114**, illustrated as a wet/dry vacuum cleaner. Although the vacuum accessory **200** is shown and described with reference to a wet/dry vacuum cleaner, the vacuum accessory **200**, the squeegee attachment assembly **216** (described further herein) and features thereof may be used with vacuum cleaners other than wet/dry vacuum cleaners including, for example and without limitation, canister vacuum cleaners, upright vacuum cleaners, and backpack vacuum cleaners.

[0038] In the example embodiment, the vacuum **114** generally includes a suction unit **102** and a vacuum conduit **104** connected in fluid communication with the suction unit **102**. The vacuum accessory **200** may be selectively connected to an end of the vacuum conduit **104**.

[0039] The suction unit **102** generally includes a motor and a fan or impeller assembly (not shown) operatively connected to the motor to drive the fan and generate suction or negative pressure differential to permit debris and other material to be collected via the conduit **104** and the vacuum accessory **200**. In the illustrated embodiment, the suction unit includes a collection drum or canister **108** and a powerhead **110** secured to the collection canister **108**. The motor and impeller assembly of the vacuum cleaning system **100** are housed within the powerhead **110** and establish a negative pressure differential or vacuum within the collection canister **108** when activated.

[0040] The suction unit **102** also includes a vacuum inlet port **112** for connection to one end of the vacuum conduit **104**. When the vacuum conduit **104** is connected to the vacuum inlet port **112**, the negative pressure differential or vacuum established by the motor and impeller assembly is transferred to the vacuum conduit **104** and creates suction along the vacuum conduit **104**. In some embodiments, the suction unit **102** may also include one or more filter or media assemblies interfaced between the vacuum inlet port **112** and the impeller assembly to collect finer particles or media entrained within the suction flow generated by the vacuum cleaning system **100**.

[0041] The suction unit **102** also includes an exhaust port (not shown) for exhausting or expelling air flow generated by the motor and impeller assembly. In some embodiments, the exhaust port may be configured for connection to the vacuum conduit **104** such that the vacuum cleaning system **100** may be used as a blower. In some embodiments, for example, the exhaust port has a configuration similar to the vacuum inlet port **112** for connection to a first end **116** of the vacuum conduit **104**.

[0042] The vacuum conduit **104** includes the first end **116** that connects to the vacuum inlet port **112** of the suction unit **102**, and a second end **118** distal from the first end **116** for connection to a vacuum cleaning accessory, such as the vacuum accessory **200**. Connection of the first end **116** of the vacuum conduit **104** to the vacuum inlet port **112** permits fluid communication between the suction unit **102** and the vacuum conduit **104** such that the negative pressure or vacuum established by the suction unit **102** creates suction along the vacuum conduit **104**. In the illustrated embodiment, the first and second ends **116** and **118** of the vacuum conduit **104** are circular in cross-section and define circular

openings for connection to the vacuum inlet port **112** and the vacuum accessory **200**, respectively.

[0043] In the illustrated embodiment, the first end **116** of the vacuum conduit **104** is releasably connectable to the vacuum inlet port **112** (e.g., by a friction fit) such that the vacuum conduit **104** may be disconnected from the suction unit **102** and stored when not in use. In other embodiments, the first end **116** of the vacuum conduit **104** may be fixed to the vacuum inlet port **112** such that the vacuum conduit **104** is not detachable from the suction unit **102**. In the illustrated embodiment, the vacuum conduit **104** includes a flexible, extendable hose. The hose may be made of a flexible material such as plastic, polypropylene (PP), polyethylene (PE), ethylene vinyl acetate (EVA), rubber, and other flexible materials. The first and second ends **116** and **118** of the vacuum conduit **104** include annular rings having a relatively rigid construction as compared to the flexible hose or conduit to facilitate connection to the vacuum inlet port **112** and the vacuum accessory **200**, respectively. In some embodiments, for example, the first end **116** and the second end **118** are constructed of the same materials as the vacuum conduit **104**, and have a more rigid construction (e.g., thicker sidewalls). In other embodiments, the ends of the vacuum conduit **104** may be constructed of any suitable semi-rigid or flexible materials that enable the vacuum cleaning system **100** to function as described herein.

[0044] In other embodiments, the vacuum conduit **104** may include a rigid tube in addition to or as an alternative to the flexible hose or conduit. In such embodiments, the tube may be constructed from suitably rigid materials including, for example and without limitation, rigid and/or pliable plastics, nylons, rubbers, and metals. In other embodiments, the vacuum conduit **104** may be constructed of any suitable material that enables the vacuum cleaning system **100** to function as described herein.

[0045] The vacuum accessory **200** is selectively connectable to the second end **118** of the vacuum conduit **104** such that the vacuum accessory **200** can be manipulated to engage surfaces **202** for cleaning (e.g., floors or other surfaces). The vacuum accessory **200** is releasably connected to the second end **118** of the vacuum conduit **104** such that the vacuum accessory **200** can be interchanged with other vacuum conduit attachment tools designed for different vacuum cleaning operations. As described in more detail herein, the vacuum accessory **200** includes a vacuum conduit connector to facilitate coupling different sized (e.g., diameter) vacuum conduits and hoses to the vacuum accessory **200**.

[0046] FIG. 2 is a perspective view of the vacuum accessory **200** that is configured for simultaneously vacuuming and squeegeeing a surface **202** (e.g., a floor). The vacuum accessory **200** includes a generally hollow upper housing or nozzle body **204** that extends from a first, downstream end **206** to a second, upstream end **208**. The nozzle body **204** defines a suction cavity **210** extending between an inlet **212** (FIG. 6) defined at the first end **206** and a vacuum outlet **214** defined at the second end **208**. The vacuum accessory **200** defines a suction flow path (generally, a flow path) extending from the inlet **212** to the vacuum outlet **214**, denoted by arrow A in FIGS. 15-20.

[0047] In reference to FIGS. 5 and 7, the vacuum accessory **200** includes a squeegee attachment assembly **216** including a lower housing **220** and a squeegee assembly **222**. The squeegee assembly **222** includes a holder assembly **224** and a squeegee blade **226** positioned in proximity to the inlet

212. The squeegee assembly **222** is rotatably connected to the lower housing **220** such that the squeegee assembly **222** is capable of rotating relative to the inlet **212**. In some alternative embodiments, the squeegee assembly **222** is rotatably connected to the nozzle body **204**. In the illustrated embodiments, the squeegee assembly **222** is rotatably connected to the lower housing **220**. In some embodiments, the nozzle body **204** is embodied as a vacuum cleaner floor tool, to which the squeegee attachment assembly **216** may be selectively connected or disconnected by a user.

[0048] The lower housing **220** is selectively connectable to the nozzle body **204**. When the lower housing **220** is connected to the nozzle body **204**, the lower housing **220** covers at least a portion of the inlet **212** defined by the nozzle body **204**, having a length L_{204} and a width W_{204} (shown in FIG. 6). In other embodiments, the lower housing **220** may be formed integrally with the nozzle body **204**.

[0049] Referring to FIGS. 5 and 7-10, the lower housing **220** includes an outer wall **227** that extends generally vertically from a base **228**. The base **228** defines an elongate, narrow suction inlet **230** having a width w_{230} defined between a front side **232** and a rear side **234**. The narrow suction inlet **230** is reduced or smaller in width than a width w_{212} of the inlet **212** defined by the nozzle body **204**. When the lower housing **220** is connected to the nozzle body **204**, the narrow suction inlet **230** may be generally centrally located between a front side **236** of the nozzle body **204** and a rear side **238** of the nozzle body **204**. The narrow suction inlet **230** extends along a length L_{230} between a first end **240** and a second end **242**, that is approximately an entire length L_{220} of the lower housing **220** and/or an entire length L_{204} of the nozzle body **204**. The squeegee assembly **222** includes a width W_{222} . The width W_{222} is less than a width W_{232} of the narrow suction inlet **230**, allowing inlet flow to pass between the squeegee assembly **222** and front and rear sides **232**, **234** of the suction inlet **230**, depending on the rotational position of the squeegee assembly **222**. See FIGS. 15-20.

[0050] The squeegee assembly **222** may be positioned generally in the middle of the narrow suction inlet **230**, between the front side **232** and the rear side **234**, separating the narrow suction inlet **230** into two sides or inlets—a forward side or inlet **246** positioned on a front side **247** of the squeegee assembly **222** and a rearward side or inlet **248** positioned on a rear side **249** of the squeegee assembly **222**. A width W_{246} of the forward inlet **246** may be the same, or generally the same, as a width W_{248} of the rearward inlet **248**.

[0051] In further reference to FIG. 5, the base **228** of the lower housing **220** includes a lower or floor-engaging surface **268** which may engage with a surface **202** when the vacuum accessory **200** is in use. In the illustrated embodiments, the lower surface **268** is serrated, having a plurality of boundary walls **267** defining a plurality of channels **269**. Each of the channels **269** are arranged or oriented perpendicular to a front side **260** and a rear side **262** of the lower housing **220**, and to the narrow suction inlet **230**. In the illustrated embodiment, the plurality of channels **269** are arranged on both sides of the narrow suction inlet **230**. A portion of the boundary wall **267** may be in contact with a surface **202**, during use of the vacuum accessory **200**, and at least a portion of the channel **269** is spaced from the surface **202**, providing an air channel leading toward the narrow suction inlet **230**.

[0052] In some embodiments, the lower housing **220** includes one or more retainers **250** for engaging a retainer wall **252** of the nozzle body **204**. See FIG. 3. The lower housing **220** may further include a connector **253** including one or more clips **254** for engaging one or more engagement features **256** on the nozzle body **204**, as shown in FIG. 4. In the illustrated embodiment, the retainers **250** include a flange **258** that extends inwardly from the outer wall **227** to define a retention channel **259**, as shown in FIGS. 15-20. During assembly, the retainer wall **252** may be inserted into the retention channel **259** to secure the lower housing **220** and squeegee attachment assembly **216** to the nozzle body **204**.

[0053] In some embodiments, the retainer **250** may be positioned on, or formed integrally with, the front side **260** of the lower housing **220**. The connector **253** may be positioned on the opposing rear side **262** of the lower housing **220**. Similarly, the retainer wall **252** may be positioned on at least the front side **236** of the nozzle body **204** and the engagement feature **256** may be positioned on the opposing rear side **238** of the nozzle body **204**. Engagement of the retainers with the retainer wall **252** and engagement of the connector **253** with the engagement feature **256** may be used to connect the lower housing **220** to the nozzle body **204**. In the illustrated embodiment, the retainer wall **252** is formed around the perimeter of the inlet **212** of the nozzle body **204**, and the retainer wall **252** and the engagement feature **256** may be formed integrally together.

[0054] In some embodiments, the clip **254** may define a recess **264**, and the engagement feature **256** may include a protrusion **266** that is sized and shaped to be received or extend into the recess **264**. When the lower housing **220** is connected to the nozzle body **204**, the clip **254** retains the protrusion **266** within the recess **264** to secure the lower housing **220** to the nozzle body **204**, as shown, for example, in FIGS. 15-20.

[0055] In some embodiments, the connector **253** further includes a biasing element **270** such that the clip **254** may be moved between an initial, at rest position **272** (shown in FIG. 4) and an extended position (not shown) in which the clip **254** is moved outwardly away from the nozzle body **204** and/or relative to the initial, inward, position **272**. In the illustrated embodiment, the biasing element **270** is embodied as a portion of the lower housing **220** that has a thickness and/or is composed of a suitable material allowing the clip **254** to flex and extend between the initial position **272** and the extended position. In other embodiments, the biasing element **270** may be embodied as a torsional spring. In the inward, or initial, position **272**, the clip **254** is connected to the one or more engagement features **256** of the nozzle body **204**. The biasing element **270** enables the squeegee attachment assembly **216** to be selectively connected or disconnected to the nozzle body **204** without the use of tools or additional fasteners.

[0056] In some embodiments, the biasing element **270** enables the clip **254** to be moved to the extended position, such that the clip **254** may be moved past the protrusion **266** such that the protrusion **266** may be positioned within the recess **264**. The connector **253** may further include a gripping portion **274** that may be pressed on by a user, to connect or disconnect the clip **254** and the engagement feature **256**. For example, the gripping portion **274** may be pressed causing the biasing element **270** to move the clip **254** from the inward position **272** to the extended position. In other

embodiments, the lower housing 220 may be selectively connected to the nozzle body 204 using any suitable attachment mechanisms or retaining features without the use of separate fasteners, e.g., bolts or screws and without the use of additional tools.

[0057] The lower housing 220 may include an attachment channel 279 defined between the outer wall 227 and at least one inner wall 278 sized to receive a portion of the nozzle body 204 therein, to selectively align and/or connect the nozzle body 204 to the lower housing 220. For example, at least a portion of the nozzle body 204, e.g., the retainer wall 252, may be pressed into the attachment channel 279, between the outer wall 227 and the at least one inner wall 278, such that the lower housing 220 may also be frictionally connected to the nozzle body 204.

[0058] In reference to FIGS. 7-10, the lower housing 220 further includes one or more clips 280 that rotationally support the squeegee assembly 222 within the lower housing 220. In the illustrated embodiment, the lower housing 220 includes three of the clips 280 positioned along the length of the narrow suction inlet 230. For example, in the illustrated embodiment, a first clip 282 is disposed at the first end 240 of the narrow suction inlet 230, a second clip 286 is disposed at the second end 242 of the narrow suction inlet 230, and a third clip 288 is positioned between the first end 240 and the second end 242.

[0059] Each clip 280 includes an arched wall 300 that extends over the narrow suction inlet 230 and defines a clip pocket 302. As described below, at least a portion of the squeegee assembly 222 is positioned within the clip pocket 302 and is rotatable relative to the arched wall 300.

[0060] In reference to FIGS. 7, 8, 11, and 14, the squeegee assembly 222 further includes one or more rotational bearing elements 310 extending from or formed integrally with, the holder assembly 224. The rotational bearing element 310 is sized and shaped to be received within the clip pocket 302 with sufficient clearance between the clip 280 and the rotational bearing element 310 such that the rotational bearing element 310 may rotate within the clip pocket 302 and relative to the clip 280. For example, the arched wall 300 may define a circular shaped clip pocket 302, and the rotational bearing element 310 may be circular shaped and sized to fit within the circular shaped clip pocket 302, enabling rotation of the squeegee assembly 222 relative to the arched wall 300.

[0061] In reference to FIG. 10, each clip 280 includes a neck 312 that defines an inlet 314 to the clip pocket 302. The inlet 314 has a width d_{312} that is narrower than a diameter d_{310} , or a width, of the rotational bearing elements 310. Moreover, the neck 312 can be formed of suitably resilient or compliant materials such that the neck 312 can flex and the width d_{312} of the inlet 314 can be increased. The rotational bearing elements 310 can thereby be pressed through the neck 312, widening the inlet 314, such that the rotational bearing element 310 may be positioned within the clip pocket 302. When the rotational bearing element 310 is disposed within the clip 280, the neck 312 restricts disengagement between the clip 280 and the rotational bearing element 310. Accordingly, the squeegee attachment assembly 216 including the lower housing 220, the holder assembly 224, including a first holder bracket 340 and a second holder bracket 342, and the squeegee blade 226 may be assembled together without the use of tools and/or separate fasteners or tools.

[0062] In reference to FIGS. 7-8, the clip 280 includes a slot 320 and the rotational bearing element 310 includes a hinge 322 extending through the slot 320. In the illustrated embodiment, two of the clips 280 (i.e., the first and second clips 280 located at the first and second ends 240, 242 of the narrow suction inlet 230) include a slot 320, and two of the corresponding rotational bearing elements 310 include a hinge 322. The hinges 322 rotationally connect the first holder bracket 340 relative to the second holder bracket 342. For example, each hinge 322 includes a thickness and/or is composed of a suitable material allowing the first holder bracket 340 and second holder bracket 342 to rotate about the hinges 322. In some alternative embodiments, the hinges 322 include a biasing element, such as a torsional spring. The hinges 322 connect the first holder bracket 340 and the second holder bracket 342. For example, the first holder bracket 340, the second holder bracket 342, and the hinges 322 are formed integrally together, improving manufacturing efficiency by reducing the number of individual components. In some alternative embodiments, the first holder bracket 340 and the second holder bracket 342 are formed separately and are connected together by one or more hinges 322.

[0063] The slot 320 is at least partially defined by a first end 324 and a second end 326. Rotation of the rotational bearing element 310 relative to the clip 280 moves the hinge 322 within the slot 320. The slot 320 may provide clearance for the hinge 322 to extend through the arched wall 300 of the clip 280.

[0064] In the illustrated embodiment, when the rotational bearing element 310 is disposed within the clip pocket 302, the hinge 322 is disposed between the first end 324 and the second end 326. In some embodiments, such as the illustrated embodiment, more than one of the clips 280 and corresponding rotational bearing elements 310 includes the slot 320 and hinge 322.

[0065] In some alternative embodiments, the hinges 322 act as stoppers, such that contact between the hinge 322 and the first end 324 or the second end 326 stops or arrests continued rotation of the rotational bearing element 310, stopping continued rotation of the squeegee assembly 222 in a given direction. The hinge 322 and slot 320 may be sized and shaped to allow for a desired amount of rotation of the squeegee assembly 222 relative to the nozzle body 204, the lower housing 220, the inlet 212, and/or the narrow suction inlet 230, as described further herein with reference to FIGS. 15-20.

[0066] In reference to FIGS. 11 and 12, the squeegee blade 226 includes a base portion 330 and a blade portion 332. The base portion 330 is connected to or received within the holder assembly 224, such that the squeegee assembly 222 and the holder assembly 224 rotate together relative to the lower housing 220, the inlet 212, and/or the narrow suction inlet 230.

[0067] In reference to FIG. 14, the first holder bracket 340 and the second holder bracket 342 are rotatable about hinges 322 between a closed position 378, see FIGS. 11, 13 and 14, and an open position (not shown). In the closed position, the first holder bracket 340 and the second holder bracket 342 are selectively connectable together, for example, by one or more connector assemblies 370. The first holder bracket 340 defines a first holder pocket 344 and the second holder bracket 342 defines a second holder pocket 346, as shown in FIGS. 15-20. The first and second holder brackets 340, 342

may be selectively connected together (e.g., connector assemblies 370) to form a single holder pocket 348 formed of the first and second holder pockets 344, 346. The holder pocket 348 is sized and shaped to receive and retain the base portion 330 therein when the first holder bracket 340 and the second holder bracket 342 are in the closed position.

[0068] The holder assembly 224 includes an outlet 350, defined by the first holder bracket 340 and the second holder bracket 342. The blade portion 332 of the squeegee blade 226 extends outside of the holder pocket 348 through the outlet 350, and beyond the base 228 of the lower housing 220.

[0069] In some embodiments, the rotational bearing element 310 may also be formed of two separate components, for example, a first side 352 connected to, or formed integrally with, the first holder bracket 340 and a second side 354 connected to, or formed integrally with, the second holder bracket 342. When the first holder bracket 340 is connected to the second holder bracket 342, e.g., in the closed position 378, the first and second sides 352, 354, are connected together to form the entire rotational bearing element 310. Additionally, in some embodiments, each hinge 322 is also formed of a first hinge side 356 connected to, or formed integrally with, the first holder bracket 340 and a second hinge side 358 that is connected to, or formed integrally with, the second holder bracket 342. The first hinge side 356 and the second hinge side 358 are hinged together at a distal end 359 of the hinge 322. In some embodiments, like the illustrated embodiment, the first holder bracket 340 and the second holder bracket 342 include a lower ridge 360, as shown in FIGS. 13. The base portion 330 of the squeegee blade 226 includes a lip 362, such that when the base portion 330 of the squeegee blade 226 is disposed within the holder pocket 348, the lip 362 is in contact with the ridge 360 to retain the base portion 330 within the holder pocket 348, e.g., the connection between the lip 362 and the ridge 360 may prevent the base portion 330 from moving through the outlet 350.

[0070] In reference to FIG. 13, the first holder bracket 340 and second holder bracket 342 may be selectively connected or disconnected together using one or more connector assemblies 370. For example, in the closed position 378, the first and second holder brackets 340, 342 may be connected together, using one or more connector assemblies 370, to prevent rotation about of the first and second holder brackets 340, 342 about the hinges 322. Each connector assembly 370 of the illustrated embodiment includes a keyed opening 372 connected to one of the first holder bracket 340 or second holder bracket 342, and at least one keyed clip 374 connected to the other of the first holder bracket 340 and second holder bracket 342. When the first and second holder brackets 340, 342 are pressed together, the keyed clip 374 may be disposed within the keyed opening 372.

[0071] In further reference to FIG. 14, the first holder bracket 340 and the second holder bracket 342 of the illustrated embodiment include a first alignment feature 380 formed on one of the first and second holder brackets 340, 342, and a second alignment feature 382 formed on the other of the first and second holder brackets 340, 342. The first and second alignment features 380, 382 are sized and shaped to align the first holder bracket 340 relative to the second holder bracket 342 when the alignment features 380, 382 are in contact. Mating engagement of the first and second alignment features 380, 382 aligns the keyed opening 372

with the keyed clip 374 to facilitate the connection of the first holder bracket 340 with the second holder bracket 342. In the illustrated embodiment, the first alignment feature 380 and the second alignment feature 382 are defined on the first side 352 and second side 354 of the rotational bearing element 310. In other embodiments, the first alignment feature 380 and second alignment feature 382 may be formed or attached to any suitable location on the first holder bracket 340 and the second holder bracket 342. In embodiments described herein, the first holder bracket 340 and the second holder bracket 342 may be selectively connected together, in the closed position 378, or disconnected, in the open position, without the use of additional fasteners, e.g., screws or bolts, and without the use of additional tools.

[0072] Referring again to FIGS. 2-6, the vacuum accessory 200 includes a vacuum connector 390, positioned near the second end 208 around the vacuum outlet 214, for connection to the vacuum, e.g., for connection to the second end 118 of the vacuum conduit 104 (or hose). The vacuum connector 390 is configured (sized, shaped, and made of suitable material) for connection to the vacuum conduit 104 and can be designed for connecting to various sizes (e.g., diameters) of vacuum hoses or conduits. When the vacuum accessory 200 is connected to the vacuum conduit 104 via the vacuum connector 390, suction generated by the suction unit 102 is transferred to the vacuum accessory 200, generating airflow through the inlet 212 and/or narrow suction inlet 230 at the first end 206 of the vacuum accessory 200 towards the second end 208 of the vacuum accessory 200.

[0073] In reference to FIGS. 15-20, the squeegee assembly 222 is rotatable between a neutral position 400 (shown in FIGS. 15 and 16), a rearward position 402 (shown in FIGS. 17 and 18), and a forward position 404 (shown in FIGS. 19 and 20). In the neutral position 400, shown in FIGS. 15 and 16, the squeegee assembly 222, e.g., the blade portion 332, is oriented generally perpendicular to the lower housing 220 and/or the narrow suction inlet 230 and is positioned approximately midway between the front side 232 and rear side 234 of the narrow suction inlet 230. In addition, in the neutral position 400, the hinge 322 is positioned halfway, or midway, between the first end 324 and the second end 326 of the slot 320. In the neutral position 400, the flow path A includes a front flow path A_f through the forward inlet 246 and a rear flow path A_r through the rearward inlet 248. See FIG. 16. In some embodiments, the squeegee assembly 222 may be positioned in the neutral position 400 as the squeegee assembly 222 is rotated between the forward position 404 and rearward position 402 and vice versa. In some embodiments, the rearward position 402 and the forward position 404 are at least 10 degrees of rotation from the neutral position. In some embodiments, for example, the rearward position 402 and the forward position 404 are approximately 15 degrees of rotation from the neutral position 400. In yet other embodiments, the rearward position 402 and the forward position 404 is between 10 degrees and 20 degrees of rotation from the neutral position 400. In some embodiments, the forward position 404 and the rearward position 402 are approximately equal degrees of rotation from the neutral position 400.

[0074] In reference to FIG. 18, when the squeegee assembly 222 is positioned in the rearward position 402, at least a portion of the squeegee assembly 222 is in contact with the rear side 234 of the narrow suction inlet 230, substantially blocking or occluding the rearward inlet 248, and leaving the

forward inlet **246** unblocked to allow air flow along flow path A_f , while preventing or limiting air flow along flow path A_r .

[0075] Contact between the squeegee assembly **222** (e.g., the holder assembly **224**) and the rear side **234** of the narrow suction inlet **230** resists further rotations of the squeegee assembly **222**. In some embodiments, the squeegee blade **226** does not restrict rotations and/or does not contact the lower housing **220**, reducing wear on the squeegee blade **226**.

[0076] In some alternative embodiments, when the squeegee assembly **222** is positioned in the rearward position **402**, the hinge **322** may act as a stopper and may be in contact with the first end **324** of the slot **320**, restricting further rotation of the squeegee assembly **222** in the rearward direction.

[0077] In reference to FIG. **20**, when the squeegee assembly **222** is positioned in the forward position **404**, at least a portion of the squeegee assembly **222** is in contact with the front side **232** of the narrow suction inlet **230**, substantially blocking or occluding the forward inlet **246**, and leaving the rearward inlet **248** unblocked to allow air flow along flow path A_r , while preventing or limiting air flow along flow path A_f .

[0078] Contact between the squeegee assembly **222** (e.g., the holder assembly **224**) and the front side **247** of the narrow suction inlet **230** resists further rotations of the squeegee assembly **222**. In some embodiments, the squeegee blade **226** does not restrict rotations and/or does not contact the lower housing **220**, reducing wear on the squeegee blade **226**.

[0079] In some embodiments, in the forward position **404**, the hinge **322** may act as a stopper and may be in contact with the second end **326** of the slot **320**, restricting further rotation of the squeegee assembly **222** in the forward direction.

[0080] In the embodiments described herein, the squeegee assembly **222**, including the squeegee blade **226**, is automatically rotated into a useable position—i.e., a position in which the squeegee assembly **222** occludes or blocks a portion of the narrow suction inlet **230** (e.g., the forward or rearward positions)—relative to the inlet **212** and/or the narrow suction inlet **230**, by moving the vacuum accessory **200** and squeegee assembly **222** in either a forward F or rearward R direction against the surface **202**. For example, when the vacuum accessory **200** is engaged with a surface, such as floor surface **202**, and moved in the forward direction F, the friction between the surface and the squeegee blade **226** creates a torsional force on the squeegee assembly **222**, rotating the squeegee assembly **222** rearwardly, blocking or occluding the rearward inlet **248** such that the majority or all airflow through the vacuum accessory **200** flows through the forward inlet **246**. Similarly, when the vacuum accessory **200** is engaged with a surface and moved in the rearward direction R, the friction between the surface and the squeegee blade **226** creates a torsional force on the squeegee assembly **222**, rotating the squeegee assembly **222** forwardly, blocking or occluding the forward inlet **246** such that the majority or all airflow through the vacuum accessory **200** flows through the rearward inlet **248**. Rotational positioning of the squeegee assembly **222**, caused by the torsional forces, reduces, or eliminates squeaking or chattering between the squeegee blade **226** and the floor surface **202**, such that the squeegee blade **226** moves smoothly across the

surface **202** improving water and debris removal while increasing the lifespan of the squeegee blade **226**.

[0081] The vacuum accessory **200** may be constructed from a variety of suitable materials depending on the intended use or application of the vacuum accessory **200**. In some embodiments, for example, the vacuum accessory **200** is constructed of a hard, rigid plastic including, for example and without limitation, polypropylene. In other embodiments, the vacuum accessory **200** may be constructed of any suitably rigid, semi-rigid, or flexible material that enables the vacuum accessory **200** to function as described herein including, for example and without limitation, PE, EVA, and rubber. In the illustrated embodiments, the vacuum accessory **200** has a two-piece construction. That is, the hollow nozzle body **204** is formed as one piece, and other portions of the vacuum accessory **200** are formed as a separate piece (e.g., an outer collar). When the two pieces are coupled together, the vacuum accessory **200** is formed, as described further herein. Suitable methods for forming individual pieces of the vacuum accessory **200** include, for example and without limitation, injection molding, precision machining, and casting.

[0082] The squeegee assembly **222**, including the base portion **330** and the blade portion **332**, may be constructed of a different material than the nozzle body **204** and/or the lower housing **220**. For example, the squeegee assembly **222** may be constructed of rubber, urethane, neoprene, and/or polyurethane. In some embodiments, the base portion **330** and the blade portion **332** are composed of different materials.

[0083] An example method of assembling the vacuum accessory **200** including the nozzle body **204** and the squeegee attachment assembly **216** includes positioning the base portion **330** of the squeegee blade **226** in the holder pocket **348** defined by one of the first holder bracket **340** and/or the second holder bracket **342** of the holder assembly **224**. The method further includes rotating the first holder bracket **340** and the second holder bracket **342** about the hinge **322** into the closed position **378** capturing the base portion **330** within the holder pocket **348**.

[0084] The method further includes connecting the first holder bracket **340** and the second holder bracket **342** together, in the closed position **378**, retaining the base portion **330** within the holder pocket **348**. The first holder bracket **340** and the second holder bracket **342** may be connected together, when rotated into the closed position **378**, by matingly engaging the first alignment feature **380** with the second alignment feature **382**, aligning the relative position of the first holder pocket **344** and second holder pocket **346** and aligning the keyed opening **372** and the keyed clip **374**. Connecting the first holder bracket **340** and the second holder bracket **342** together may also include inserting the keyed clip **374** into the keyed opening **372**.

[0085] The method also includes rotatably connecting the holder assembly **224** to the clip **280** of the lower housing **220** such that the squeegee assembly **222** is at least partially disposed within the narrow suction inlet **230** and rotatable relative to the narrow suction inlet **230** between the forward position **404**, in which the squeegee assembly **222** occludes the forward inlet **246** of the narrow suction inlet **230**, and the rearward position **402**, in which the squeegee assembly **222** occludes the rearward inlet **248** of the narrow suction inlet **230**.

[0086] The method also includes connecting the squeegee attachment assembly 216 to the nozzle body 204. Connecting the squeegee attachment assembly 216 to the nozzle body 204 can include inserting at least a portion of the nozzle body 204 in the attachment channel 279 defined between the outer wall 227 and the one or more inner walls 278. Connecting the squeegee attachment assembly 216 to the nozzle body 204 can also include inserting the retainer wall 252 into the retention channel 259 to connect the front side 236 of the lower housing 220 to the front side 236 of the nozzle body 204. Connecting the squeegee attachment assembly 216 to the nozzle body 204 can also include engaging the clip 254 of the lower housing 220 with the engagement feature 256 on the nozzle body 204.

[0087] Example embodiments of squeegee attachment assemblies are described above in detail. Aspects of the squeegee attachment assemblies are not limited to the specific embodiments described herein, but rather, components of the squeegee attachment assemblies may be used independently and separately from other components described herein. For example, in some embodiments, the lower housing may be formed separately from the nozzle body and may be selectively connected or disconnected from the nozzle body. In alternative embodiments, the lower housing may be formed integrally with the nozzle body.

[0088] As used herein, the terms “about,” “substantially,” “essentially” and “approximately” when used in conjunction with ranges of dimensions, concentrations, temperatures or other physical or chemical properties or characteristics is meant to cover variations that may exist in the upper and/or lower limits of the ranges of the properties or characteristics, including, for example, variations resulting from rounding, measurement methodology or other statistical variation.

[0089] When introducing elements of the present disclosure or the embodiment(s) thereof, the articles “a,” “an,” “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” “containing” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The use of terms indicating a particular orientation (e.g., “top,” “bottom,” “side”, etc.) is for convenience of description and does not require any particular orientation of the item described.

[0090] As various changes could be made in the above constructions and methods without departing from the scope of the disclosure, it is intended that all matter contained in the above description and shown in the accompanying drawing[s] shall be interpreted as illustrative and not in a limiting sense.

What is claimed is:

1. A vacuum accessory comprising:

- a nozzle body defining a cavity and a vacuum inlet port selectively connectable to a vacuum conduit; and
- a squeegee attachment assembly removably connected to the nozzle body, the squeegee attachment assembly comprising:
 - a housing selectively connectable to the nozzle body, wherein the housing defines a suction inlet; and
 - a squeegee assembly rotatably connected to the housing such that the squeegee assembly is rotatable relative to the suction inlet, wherein the squeegee assembly is at least partially disposed within the suction inlet, wherein the squeegee assembly is rotatable between a forward position, in which the squeegee assembly

occludes a forward side of the suction inlet, and a rearward position, in which the squeegee assembly occludes a rearward side of the suction inlet.

2. The vacuum accessory of claim 1, wherein the housing includes a lower surface defining a plurality of channels oriented perpendicular to the suction inlet.

3. The vacuum accessory of claim 1, wherein the squeegee assembly is rotatable from a neutral position to the forward position and from the neutral position to the rearward position, wherein the forward position and the rearward position are approximately equal degrees of rotation from the neutral position.

4. The vacuum accessory of claim 3, wherein the forward position and the rearward position are at least 10 degrees of rotation from the neutral position.

5. The vacuum accessory of claim 1, wherein the squeegee assembly comprises:

- a holder assembly including a first holder bracket and a second holder bracket rotatably connected to the first holder bracket, wherein the first and second holder brackets are rotatable between an open position and a closed position in which the holder assembly defines a holder pocket; and
- a squeegee blade including a blade portion and a base portion, wherein the base portion is sized and shaped to be received within the holder pocket and the blade portion extends outside of the holder pocket.

6. The vacuum accessory of claim 5, wherein one of the first holder bracket and the second holder bracket includes a keyed opening and the other of the first holder bracket and the second holder bracket includes a keyed clip sized and shaped to be received within the keyed opening, wherein the keyed clip is disposed within the keyed opening to selectively connect the first holder bracket to the second holder bracket.

7. The vacuum accessory of claim 5, wherein the first holder bracket defines a first alignment feature and the second holder bracket defines a second alignment feature, wherein the first and second alignment features are in contact with one another when the first holder bracket and the second holder bracket are in the closed position.

8. The vacuum accessory of claim 5, wherein the holder assembly contacts a front side of the suction inlet when the holder assembly is rotated in the forward position, thereby restricting further forward rotation, and wherein the holder assembly contacts a rear side of the suction inlet when the holder assembly is rotated in the rearward position, thereby restricting further rearward rotation.

9. The vacuum accessory of claim 1, wherein the housing includes a clip positioned over the suction inlet and defining a clip pocket, wherein at least a portion of the squeegee assembly is disposed within the clip pocket, and wherein the squeegee assembly is rotatable within the clip pocket between the forward and rearward positions.

10. The vacuum accessory of claim 9, wherein the housing includes a plurality of clips disposed at different locations along a length of the suction inlet.

11. A squeegee attachment assembly for use with a vacuum accessory, the squeegee attachment assembly comprising:

- a housing defining a suction inlet, the housing including at least one connector to selectively connect the squeegee attachment assembly to a nozzle body; and

a squeegee assembly rotatably connected to the housing such that the squeegee assembly is rotatable relative to the suction inlet, wherein the squeegee assembly is at least partially disposed within the suction inlet, wherein the squeegee assembly is rotatable between a forward position, in which the squeegee assembly occludes a forward side of the suction inlet, and a rearward position, in which the squeegee assembly occludes a rearward side of the suction inlet.

12. The squeegee attachment assembly according to claim **11**, wherein the housing includes a lower surface defining a plurality of channels oriented perpendicular to the suction inlet.

13. The squeegee attachment assembly according to claim **11**, wherein the squeegee assembly is rotatable from a neutral position to the forward position and from the neutral position to the rearward position, wherein the forward position and the rearward position are approximately equal degrees of rotation from the neutral position.

14. The squeegee attachment assembly according to claim **13**, wherein the forward position and the rearward position are at least 10 degrees of rotation from the neutral position.

15. The squeegee attachment assembly according to claim **11**, wherein the squeegee assembly comprises:

- a holder assembly including a first holder bracket and a second holder bracket rotatably connected to the first holder bracket, wherein the first and second holder brackets are rotatable between an open position and a closed position in which the holder assembly defines a holder pocket; and

- a squeegee blade including a blade portion and a base portion, wherein the base portion is sized and shaped to be received within the holder pocket and the blade portion extends outside of the holder pocket.

16. The squeegee attachment assembly according to claim **15**, wherein one of the first holder bracket and the second holder bracket includes a keyed opening and the other of the first holder bracket and the second holder bracket includes a keyed clip sized and shaped to be received within the keyed opening, wherein the keyed clip is disposed within the keyed opening to selectively connect the first holder bracket to the second holder bracket.

17. The squeegee attachment assembly according to claim **15**, wherein the first holder bracket defines a first alignment feature, and the second holder bracket defines a second alignment feature, wherein the first and second alignment features are in contact with one another when the first holder bracket and the second holder bracket are in the closed position.

18. The squeegee attachment assembly according to claim **15**, wherein the holder assembly contacts a front side of the suction inlet when the holder assembly is rotated in the forward position, thereby restricting further forward rotation, and wherein the holder assembly contacts a rear side of the suction inlet when the holder assembly is rotated in the rearward position, thereby restricting further rearward rotation.

19. The squeegee attachment assembly according to claim **11**, wherein the housing includes a clip positioned over the suction inlet and defining a clip pocket, wherein at least a portion of the squeegee assembly is disposed within the clip pocket, and wherein the squeegee assembly is rotatable within the clip pocket between the forward and rearward positions.

20. A method of assembling a vacuum accessory including a nozzle body and a squeegee attachment assembly, the squeegee attachment assembly including a squeegee assembly and a housing defining a suction inlet, the method comprising:

- positioning a base portion of a squeegee blade in a holder pocket defined by one of a first holder bracket and a second holder bracket of a holder assembly;

- connecting the first holder bracket to the second holder bracket to retain the base portion within the holder assembly;

- rotatably connecting the holder assembly to a clip of the housing such that the squeegee assembly is at least partially disposed within the suction inlet and rotatable relative to the suction inlet between a forward position, in which the squeegee assembly occludes a front side of the suction inlet, and a rearward position, in which the squeegee assembly occludes a rear side of the suction inlet; and

- connecting the squeegee assembly to the nozzle body.

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